



User Guide

Alif Security Toolkit

V1.112.0

Table of Contents

Introduction	6
1. Python Executables	7
2. Source code release	7
3. Release Version	7
4. Quick User Guide	8
Updating the latest ALIF release	8
Adding / Changing an application image	8
5. TOC Packages	10
System Table of Contents (STOC)	10
SE-UART Output Fields	11
SEROM	11
Banner	11
LCS (Life Cycle State)	11
[SES] Output	11
Device configuration	11
[SES] Table	11
SES Boot Error messages	13
Firewall Exception messages	13
Firewall Configuration Errors	14
Error messages – no ATOC in MRAM	14
Application Table of Contents (ATOC)	14
ATOC Placed Images	15
ATOC Policies	15
ATOC Booting	15
ATOC Part Number Mismatch	15
NTOC Images	16
6. Alternate SES Boot Path Option	17
Envisaged Use case for Alternate Boot Path	17
Alternate Boot Path	18
Definition of MRAM “bad”?	19

Recovery Mode Support ENSEMBLE REV_A0/REV_A1	19
SETOOLS Support	20
SETOOL Package Support.....	20
Provisioning Alternate Boot Path.....	20
Provisioning Details.....	20
Provisioning Details BOOTOPT	21
Provisioning Details OTPOSPISettings	21
Provisioning Steps	21
7. ISP (In System Programming).....	22
SE-UART Connection	22
UART Settings.....	22
Dynamic Baud Rate Change	22
Baud Rate Override	22
UART Errors – PermissionError	22
UART Errors – Disconnected	23
UART Errors – FileNotFoundError	23
ISP Discovery	23
ISP when in Low power mode.....	24
8. Tool Flow.....	25
Debug Stubs	25
MRAM Burners.....	26
9. Running the Tools	27
tools-config tool	27
Tools-config command line mode	28
Discovery	29
Command Port	29
Baud Rate	30
app-gen-rot tool.....	31
app-gen-toc tool	33
External OSPI memory support.....	37
Valid LoadAddress Regions	38
Note on Encryption and XIP	38
Note on Factory (Blinky) for ENSEMBLE E1C and BALLETO B1	38
app-write-mram tool	40
Maintenance Mode Internals	41

Maintenance Mode reporting.....	41
Erase option	41
External OSPI Memory Boot support.....	42
Padding binaries (-p option)	43
No Reset option (-nr)	43
Images option (-i)	43
SKIP Images option (-S)	44
SWITCH option (-s)	44
No reset option (-nr)	44
COMPORT (-c)	44
ATOC Policy	46
Part Number and/or Part Revision mismatch warnings	46
Using app-write-mram in SEROM Recovery mode	47
UpdateSystemPackage tool	48
No Reset option (-nr)	48
No Authentication (-na) option.....	49
Memory option (-m)	49
Setting the target device as the default.....	51
Maintenance tool.....	52
Running the Tool.....	52
Command line options (-opt).....	53
BAUD option (-b).....	56
COMPORT option (-c).....	57
Maintenance Menu Grouping.....	58
maintenance menu in SEROM Recovery mode	59
Hard maintenance Mode	59
Soft maintenance Mode	59
Device Reset.....	59
Device Information	60
MRAM	79
Utilities	81
Setting capabilities	85
ROM	86
app-assets-gen	91
Procedure to encrypt assets	93

app-provision	95
app-secure-debug (--rma)	96
Tool Exit codes	98
NACK Error Handling	99
TIMEOUT Error Handling.....	99
10. RECOVERY MODE	100
SEROM Error Codes.....	100
11. Primary Boot Path Alternative	102
12. CONDUCTOR Tool Flow.....	103
Configuration Options.....	103
STATIC configuration.....	103
DYNAMIC configuration	103
Using the CONDUCTOR Tool with SETOOLS.....	103
Miscellaneous Feature set supported from CONDUCTOR tool	104
SE_BOOT_INFO	105
Non-CONDUCTOR Options.....	105
Brown Out Reset (BOR) hysteresis and threshold configurable	105
SHARED_MEM_CTRL.....	106
REG_ALIASING_CTRL.....	106
SE_CLK_SRC_PLL	107
Open Firewall configuration	107
13. Device Debug Strategies	108
Target Not responding	108
Recovery vs SES Mode	108
HARD Maintenance mode	108
Quick Fix guide	109
Reporting Issues to ALIF.....	109
14. Life Cycle State SECURE	110
15. Document History.....	111

Introduction

The Alif Security Toolkit (SETOOLS) contains several tools (written in Python3) required to program and provision an Alif Semiconductor SoC device.

Provisioning means being able to add images to the internal NVM storage (MRAM) as well as adding Security assets such as Keys to the One Time Programmable memory (OTP).

ALIF images, known as System TOC (STOC), for the device are already provisioned into the MRAM, a user cannot change these, but can update to a new STOC version supplied by Alif. User images such as Application programs (binaries written for A32 or M55 cores) need to be packaged and written to the MRAM. The packages are called an Application Table of Contents (ATOC).

There are also pre-built binary images for the ALIF Table of Contents (STOC) Package. STOC contains the latest binary version of SERAM. The STOC also contains debug stubs which will execute on the Application cores to allow connection from the Debugger. These stubs are only loaded to Application CPUs that are not configured to run.

NOTE: These tools are designed to work with the following environment:

- Windows PC Executables (Windows 10/11)
- LINUX Executables (Ubuntu 18.4/20.4/22.4)
- MacOS Executables (tested with OS Sonoma 14.1).
- Ensemble DevKit Gen 2

The Alif [Ensemble DevKit User Guide \(Gen 2\)](#) includes instructions on how to connect the SE-UART output of the CPU to the host computer. This guide will then refer to the [Alif Security Toolkit Quick Start Guide](#) which contains instructions for installation of these tools.

This release version is **`app-release-exec-windows(linux)(macos)-SE_FW_1.110.0_DEV.zip`**

1. Python Executables

The Security tools are executable images, previously these python scripts were delivered as source code. These executables do not require python or any pre-requisite libraries to be installed.

Example:

```
$ ./maintenance
[INFO] COM8 open Serial port success
[INFO] baud rate 55000

Available options:

1 - Device Control
2 - Device Information
3 - MRAM
4 - Utilities
5 - Setting capabilities
6 - ROM

Select an option (Enter to exit): |
```

Previously you were required to run

```
$ python3 <tool-name>.py
```

Now you just run

```
$ <tool-name>
```

```
$ ./<tool-name>
```

NOTE: Screenshots in this document and other application notes or user guides might show the previous format but apply to this version or later releases if you enter just <tool-name> instead of “python3 <tool-name>.py”.

2. Source code release

The SETOOLS source code release requires the user to install some supporting python packages

```
$ pip install pyqt5
$ pip install cryptography
$ pip install pyserial
$ pip install pylink-square
```

3. Release Version

Inside the SETOOLS app release directory there is

- An empty text file denoting the version of *this* release.

```
SETOOLS_version_SE_FW_1.112.00
```

- A file version.txt

- This contains the version e.g. V1.112.0

4. Quick User Guide

Please ensure you have followed the Installation Guide in the Alif Security Toolkit Quick Start Guide before proceeding.

The recommended <release-location> installation directories are:

Windows:	C:\app-release-exec
Linux:	/home/\$USER/app-release-exec-linux
MAC OS:	/Users/\$USER/app-release-exec-macos

Updating the latest ALIF release

```
$ cd <release-location>
$ updateSystemPackage
```

```
Burning: System Package in MRAM
Selected Device:
Part# E8 (AE822FA0E5597LS0) - 5.5 MRAM / 9.75 SRAM - Rev: A0

Connecting to the target device...
[INFO] baud rate 57600
[INFO] dynamic baud rate change Enabled
[INFO] COM10 open Serial port success
Bootloader stage: SERAM
[INFO] Detected Device:
Part# AE822FA0E5597LS0 - Rev: A0
- MRAM Base Address: 0x80580000
Maintenance Mode = Enabled
Authenticate Image: True
Verify Certificate
Signature File: alif\SP-AE822FA0E5597LS0-rev-a0-dev.bin.sign
Download Image
alif\SP-AE822FA0E5597LS0-rev-a0-dev.bin[#####]100%: 270400/270400 bytes
Verify Image
Done
6.70 seconds

Authenticate Image: True
Verify Certificate
Signature File: alif\offset-58-rev-a0-dev.bin.sign
Download Image
alif\offset-58-rev-a0-dev.bin [#####]100%: 16/16 bytes
Verify Image
Done
0.02 seconds
```

Adding / Changing an application image

To change or update an image to add to MRAM do the following:

```
$ cd <app-release-location>
$ edit build/config/app-cfg.json and add new binaries images
$ app-gen-toc
```

(Or use below command to use a different configuration)

```
$ app-gen-toc -f build/config/app-myfile.json

$ app-write-mram
```


The app-gen-toc tool performs the packaging of your binary ready for writing to the MRAM.

The app-write-mram.py tool performs the writing of the application package to the MRAM. After this is completed, it will reset your target platform and the contents of the MRAM will be executed.

5. TOC Packages

The device will arrive already provisioned. This means that the SES is already programmed into MRAM.

System Table of Contents (STOC)

On the initial power up of the device you will see the following SE-UART output:

```
COM12 - Tera Term VT
File Edit Setup Control Window Help

SEROM v1.105.65 0x000002A0

SES A0 v1.112.0 Aug 3 2026 17:29:54
[SES] No ATOC
[SES] STOC DEVICE ok
[SES] No LF XTAL
[SES] Bank 0x0 marked booted
[SES] SERAM bank 0x0 is valid and booted
[SES] STOC ok
[SES] M55-HE booted from address 0x58000000
[SES] LCS=1
[SES] FC:Rgn
0:2 7:0 8:0 9:0 10:0 13:0 13:1 13:2 13:3
```

Name	CPU	Store Addr	Obj Addr	Dest Addr	Boot Addr	Size	Version	Flags	Time (ms)
DEVICE	CM0+	0x805E0EC0	0x805E04C0	-----	-----	372	0.5.0	u U	20.79
* SERAM0	CM0+	-----	0x000000C0	-----	-----	150592	1.112.0	-----	0.00
SERAM1	CM0+	-----	0x000302C0	-----	-----	150592	1.112.0	-----	0.00

```
Legend: <u><C>ompressed, <L>oaded, <U>erified, <s>kipped verification, <B>ooted, <E>ncrypted, <D>eferred

[SES] SE frequency is 77.78 MHz
```

SE-UART Output Fields

The output from the SE-UART shows what SES has processed during the boot sequence of the device.

SEROM

Shows the version of the SEROM software and the SoC revision register contents.

Banner

Shows the version of the SES software.

LCS (Life Cycle State)

Shows the LCS state of the device.

[SES] Output

Shows the state of processing the STOC and ATOC images found in MRAM.

Device configuration

```
[SES] No ATOC
[SES] System partition address 0x80580000
[SES] STOC DEVICE ok
[SES] No LP XTAL
[SES] STOC ok
```

Other images processing –

```
[SES] STOC ok
[SES] ATOC ok
```

[SES] Table

Shows the details of all TOC objects processed from MRAM.

Name	ASCII string name from the JSON file.
- DEVICE	Shows a DEVICE Config object.
- SERAM0/1	Shows the two Banks of SES software. An asterisk (*) denotes which Bank was booted from.
- HE_DBG	Shows the default Debug stub executed if no ATOC is present.
CPU	Which CPU is being targeted.
Store Address	Where in MRAM is this object stored.
Obj Address	TOC address
Dest Addr	Where in RAM is this object being copied to.
Boot Addr	What address is being used to Boot from.
Size	Size of the binary object
Version	Version ID specified in the JSON file.
Flags	
C	Image is Compressed
u	Image is Uncompressed

L	Image is LOADED to a RAM location
V	Image is VERIFIED
s	Image is SKIPPED
B	Image has been BOOTED
D	Image is DEFERRED, Booting will happen later
Time	Time in microseconds from the start of TOC processing until completion, this could include booting.

Deferred images will not display their details in the initial SES Banner print. The reason is that SES SKIPS these images on boot and so there are no details. Once the object is Un deferred then the details can be displayed.

SES Boot Error messages

SES can report the following Boot loader messages:

BL_STATUS_OK	0x00
BL_ERROR_APP_INVALID_TOC_ADDRESS	0x01
BL_ERROR_APP_INVALID_TOC_OFFSET	0x02
BL_ERROR_UNALIGNED_ADDRESS	0x03
BL_ERROR_INVALID_TOC_ADDRESS_RANGE	0x04
BL_ERROR_INVALID_TOC_FLAGS	0x05
BL_ERROR_INVALID_IMAGE_DATA	0x06
BL_ERROR_CERTIFICATE_NO_VERIFY_IN_MEMORY	0x07
BL_ERROR_CERTIFICATE_NO_VERIFY_IN_FLASH	0x08
BL_ERROR_CERTIFICATE_INVALID_LOAD_ADDRESS	0x09
BL_ERROR_CERTIFICATE_INVALID_CHAIN	0x0A
BL_ERROR_CERTIFICATE_STORAGE_ADDRESS_INVALID	0x0B
BL_ERROR_DEVICE_ADDRESS_INVALID	0x0C
BL_ERROR_UNCOMPRESS_FAILED	0x0D
BL_ERROR_SIGNATURE_VERIFY_FAILED	0x0E
BL_ERROR_APP_ACCESSING_PROTECTED_AREA	0x0F
BL_ERROR_ICV_ACCESSING_PROTECTED_AREA	0x10
BL_ERROR_FAILED_TOC_CRC32	0x11
BL_ERROR_INVALID_TOC	0x12
BL_ERROR_EXCEED_MAXIMUM_TOC_ENTRIES	0x13
BL_ERROR_NOT_IMAGE_NOT_DEVICE_CFG	0x14
BL_ERROR_INVALID_TOC_ENTRY_ID	0x15
BL_ERROR_INVALID_CPU_ID	0x16
BL_ERROR_ENTRY_NOT_SIGNED	0x17
BL_ERROR_LOAD_TO_MRAM_NOT_ALLOWED	0x18
BL_ERROR_NO_FREE_SLOTS	0x19
BL_ERROR_INVALID_M55_BOOT_ADDRESS	0x1A
BL_TOC_OBJECT_NOT_FOUND	0x1B
BL_TOC_OBJECT_FOR_CPU_NOT_FOUND	0x1C
BL_TOC_IMAGE_NOT_BOOTABLE	0x1D
BL_TOC_IMAGE_NOT_FOUND	0x1E
BL_ERROR_COMPRESSION_NOT_SUPPORTED	0x1F
BL_TOC_IMAGE_DEVICE_MISMATCH	0x20
BL_ERROR_UPD_SIGNATURE_INCORRECT	0x21
BL_ERROR_UPD_IMG_IN_MRAM_NOT_SUPPORTED	0x22
BL_ERROR_HAVE_FAILED_ATOC_IMAGES	0x23
BL_ERROR_UPD_IMG_IN_EXT_MEM_NOT_SUPPORTED	0x24

Firewall Exception messages

Firewall Components can raise exceptions at run time when they reject bus transactions because they didn't pass the various checks. Such exceptions are reported on the SES output screen.

Firewall exception from FC13, caused by Master ID 0x11 at address 0x80580000, transaction properties 0x00020000

Meaning:

FC <number> Firewall component which generated the exception.

Master ID The Corstone Master ID that issued the transaction causing the exception. It can be an APP CPU, but can also be another Corstone master like a DMA controller, USB host, etc.

Address The address targeted by the transaction.

Transaction properties are the value of the Corstone register FE_TP. The bits in it indicate the transaction properties read/write, data/instruction, unprivileged/privileged, and secure/non-secure.

Firewall Configuration Errors

On Boot up you may see messages such as:

```
[WARNING] ATOC firewall configuration skipped: FC:11 Rgn:1 error=3
```

This indicates that some firewall region configuration fragments in the ATOC device configuration files was rejected for some reason. The error code indicates the reason.

Error	Definition	Description
1	FW_CFG_ERR_MASTER_ID	The region uses the Secure Enclave's Master ID (0), which is not allowed
2	FW_CFG_ERR_ICV_CONFIG	The region is configured by the SE firmware and cannot be reconfigured
3	FW_CFG_ERR_ICV_PROT	The region overlaps with an Alif protected area (partially or fully)
4	FW_CFG_ERR_REGION_SIZE	Invalid region size code. The maximum region size code is 32 (4GB)

Error messages – no ATOC in MRAM

If there is no ATOC in MRAM, the following error messages are displayed -

```
[SES] No ATOC
[SES] System partition address 0x80580000
[SES] STOC DEVICE ok
[SES] No LF XTAL
[SES] STOC ok
[SES] M55-HE booted from address 0x58000000
[SES] LCS=1
[SES] FC:Rgn
0:2 7:1 8:1 9:1 13:0 13:1 13:2
```

Any other errors during the boot process will be reported.

Application Table of Contents (ATOC)

The default Application TOC package contains a Factory image, which Blinks the LEDs on the Development board.

Our releases consist of:

- M55_HE_TCM LED Blinky program.

This code is stored in MRAM and copied to a RAM location (The TCM for the M55_HE_CPU) and booted.

To try the Blinky program

```
$ cd <release-location>
$ app-write-mram
```

To rebuild this ATOC package

```
$ cd <release-location>
$ app-gen-toc -f build/config/app-cfg.json
$ app-write-mram
```

NOTE: When using an E1C or B1C device you must use the app-cfg-1c.json file.

ATOC Placed Images

ATOC packages allow for the absolute placement of user images in MRAM. See “mramAddress” option in the app-gen-toc configuration file.

Users must ensure that these images conform to the MRAM alignment rules and always be 16-byte aligned.

Images that are managed by the ATOC tool i.e., not placed at an absolute address are guaranteed to be aligned correctly.

ATOC Policies

Some rules applied to ATOC processing

ATOC Booting

No APPLICATION core(s) will be booted; in case the verification of an image fails. This policy applies to all Life Cycle States (LCS).

Original policy was to verify all non-bootable images in the ATOC in advance, before processing the ATOC. However, that would lead to double verification of those images (and double loading for loadable images), which would add to boot times. So, the implemented logic works as part of the regular ATOC processing. If any non-bootable image fails verification - a flag for that is set, and after that the booting of APP cores is skipped.

ATOC Part Number Mismatch

When an ATOC object is created (using app-gen-toc) the part number, set by the tools-config, is embedded within the ATOC binary.

On boot up, SES will check this part number in the ATOC image against the actual SoC Device part number. If there is a mismatch, then the ATOC object will be skipped i.e. not booted.

There are more details on this in the app-write-mram section

A note on Policy

This feature of adding the build environment Part number into the ATOC image was introduced in V109 SETOOLS. ATOC binaries built with earlier SETOOLS will not follow this Part number checking policy and will be launched by SES.

NTOC Images

NTOC (No TOC) images are Execute in place (XIP) images stored at the base address of MRAM.

M55_HE is the *only* CPU that executes these images.

These images are not generated by the Security Toolkit i.e., they are not ATOC images as they have restrictions such as they cannot be signed.

SES uses the following algorithm to process these objects:

```
IF ATOC is present
    Process ATOC and Boot
ELSE
    IF MTOC is present
        Process MTOC and Boot
    ELSE
        IF (0x80000000 and 0x80000004 has valid $SP and $PC)
            ReleaseM55_HE
        ELSE
            Load STOC Debug stub (if present)
```

SES reads the first two 32-bits of MRAM:

- Word1 = \$SP (Stack Pointer)
- Word2 = \$PC (Program Counter)

Then uses the following logic:

- Is the Program counter in the valid Application address range of MRAM?
- Is the Stack Pointer in a valid RAM memory range?

If these above criteria are not met, then the image will not be released (booted).

NOTE: MTOC support was dropped in CMSIS release V0.9.1.

6. Alternate SES Boot Path Option

ENSEMBLE devices (E8/E6/E4) support the Alternate Boot Option which means SES can be booted from External Memory instead of MRAM.

MRAM is the primary default boot path on ALIF ENSEMBLE (and BALLETTTO) SoC devices, but users can choose (via OTP provisioning) to have a Mirror (backup) boot path from external OSPI memory should MRAM become nonfunctional. This Mirror contains the STOC (ALIF) images as well as User Application images (ATOC) images.

SEROM will process flags contained in OTP denoting if this feature is enabled. These flags are provisioned by the user (as they are OTP this is only done *ONCE*) using the Maintenance tool.

Envisaged Use case for Alternate Boot Path

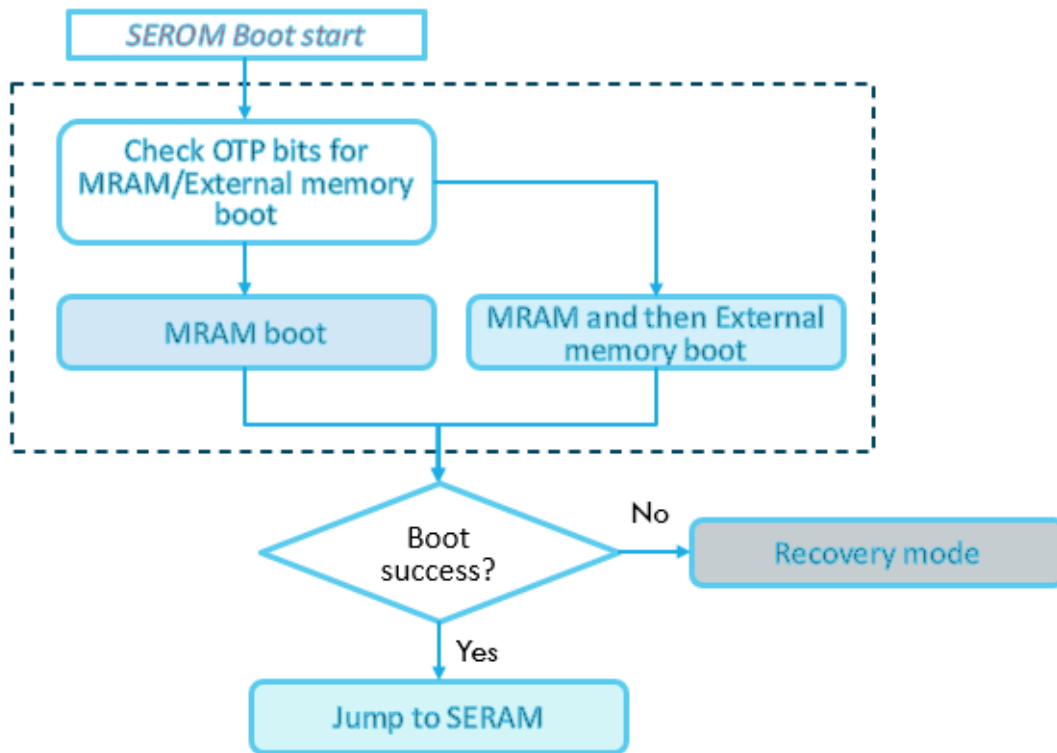
The main boot path will always be MRAM. External OSPI is considered as an *emergency backup* should MRAM fail.

When a user updates the version of SES (using UpdateSystemPackage) a similar operation using the '-m' option would be performed to update the OSPI image as well if required.

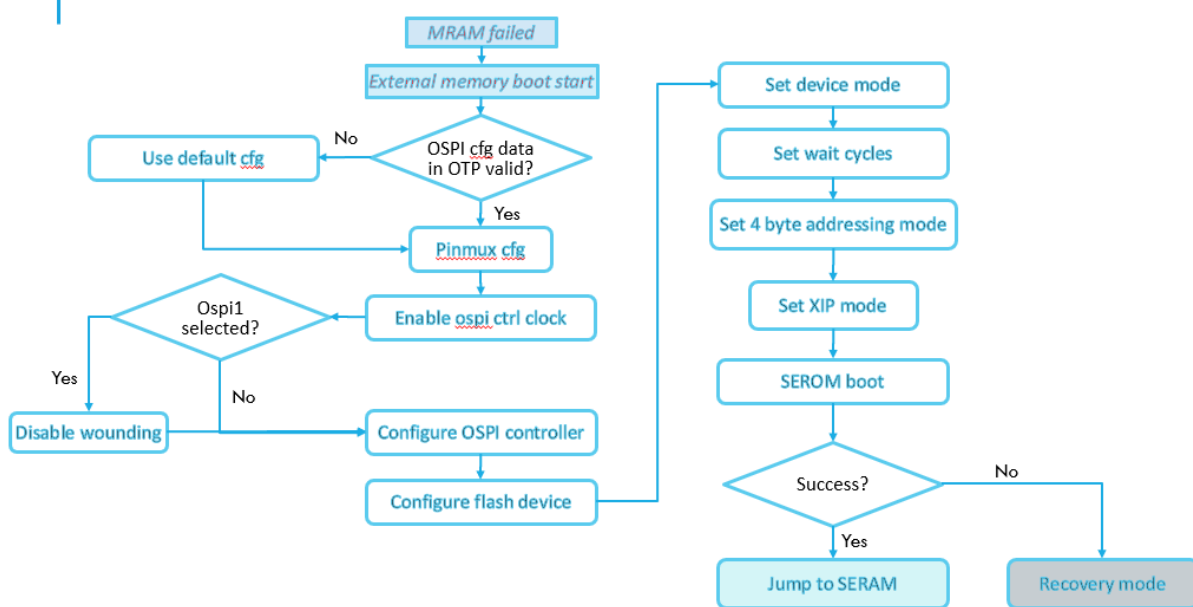
In a product, the Application would be required to update both the MRAM and External Memory (if enabled).

Alternate Boot Path

The following is the SEROM boot path showing the External memory selection path



IF the External memory path is selected then the following is executed



SETTOOLS supports the programming of the OTP as well as loading to the External OSPI memory.

Definition of MRAM “bad”?

If the MRAM contents are not correct on boot up and the External OSPI memory is selected, then SEROM will boot SES from External OSPI memory. The boot path steps are the same as from MRAM.

The definition of “bad” MRAM as seen by SEROM are:

- Failure to read offset value from the top of MRAM
 - Offset is missing i.e. 0x0
 - Offset is incorrect i.e. does not meet the bounds checks
- No STOC image in MRAM
- Failure to verify the SES image
 - Any corruption of the SES image in MRAM will result in a verification failure reported by the Cryptography engine.
 - SES image is not signed for this ALIF device
 - **NOTE:** This is unlikely to happen as SETTOOLS will not allow SES images for the incorrect device to be loaded.

Recovery Mode Support ENSEMBLE REV_A0/REV_A1

The ENSEMBLE REV_A0 E4, E6 and E8 devices do not support Recovery mode to External OSPI memory. The REV_A1 devices do support ROM based recovery to External memory.

For REV_A0 you must use UpdateSystemPackage with the option for external memory to get the STOC into the external OSPI. **NOTE:** This operation needs to be done with SES running on the device.

SETOOLS Support

SETOOLS has support for Alternate Boot path devices.

Tool	Purpose
maintenance	Shows OTP settings for Ext Memory support
maintenance	Recovery Mode to Ext Memory
UpdateSystemPackage	Allows STOC updates to Ext Memory and MRAM
app-write-mram	Allows Writing to external memory
app-gen-toc	Generate ATOCs for Ext Memory

The following is the SEROM boot path showing the External memory selection path

SETOOL Package Support

External Memory packages are available for 32M devices for OSPI0 and OSPI1 interfaces.

Provisioning Alternate Boot Path

To enable the use of External Memory support it is required to store information in OTP especially the details of the OSPI device and its configuration.

This information must be in OTP as the NVM (MRAM) can be compromised.

Provisioning Details

The OSPI configuration is encapsulated within a JSON description as follows:

```
{
  "BOOTOPT" : {
    "load" : "MRAM_THEN_EXTERNAL",
    "type" : "OSPI_FLASH_8BIT_XIP_MODE"
  },
  "OTPOSPISettings" : {
    "rx_sample_delay" : 0,
    "xip_incr_inst" : 11,
    "xip_wrap_inst" : 11,
    "aes_rxdelay" : 11,

    "ddr_drive_edge" : 1,
    "pinmux_sel" : 5,
    "bus_speed_sel" : 0,
    "fast_read_wait_cycles" : 8,
    "flash_size" : 0,
    "valid" : 1,
    "reset_port" : 15,
    "reset_pin" : 5,
    "spi_mode" : 0,

    "io_mode" : 2,
    "ddr_en" : 0,
    "xip_ddr_en" : 1,
  }
}
```

```

"inst_ddr_en"      : 0,
"xip_inst_ddr_en" : 0,
"rxds_en"         : 0,
"rxds_vl_en"      : 1,
"xip_rxds_vl_en"  : 1,
"dfs"             : 15,
"xip_dfs"         : 15,
"xip_rxds_en"     : 1,
"xip_instr_len"   : 2,
"xip_addr_len"    : 8,
"xip_spi_mode"    : 3,
"xip_hyperbus_en" : 0,
"tx_wait_count"   : 0,
"reserved_0"      : 0,

"slave_select"    : 0
}
}

```

This JSON file is stored in the `app-release/build/config/otp-altboot.json`.

There are two sections to this JSON file

- BOOTOPT
- OTPOSPISETTINGS

Provisioning Details BOOTOPT

The Boot option determines the SEROM boot path when it boots. The default is MRAM only.

Supported `load` options are

MRAM_ONLY	
MRAM_THEN_EXTERNAL	
EXTERNAL_MEM_ONLY	Not supported

Supported `type` options are

OSPI_FLASH_8BIT_XIP_MODE	
OSPI_FLASH_8BIT_SPI_MODE	
NAND_FLASH_8BIT_BURST_MODE	
SPI_FLASH_1BIT_STD_MODE	

Provisioning Details OTPOSPISETTINGS

The OTP OSPI Settings option contains the OSPI specific device details. These details are used by SEROM and SES to initialize the OSPI driver to enable Erase, Read and Write operations.

Provisioning Steps

To enable programming of the OTP, use the Maintenance tool -> Utilities -> Alternate Boot OTP (See Maintenance Tool section for details).

7. ISP (In System Programming)

ISP is a mechanism for connecting to the Secure Enclave using the Secure Enclave UART (SE-UART) and performing various operations with the device.

Please ensure the prerequisite python packages are installed before starting. See the installation instructions in the Alif Security Toolkit Quick Start Guide.

SE-UART Connection

UART Settings

Baud Rate	57600 (default)
Data	8 bits
Parity	None
Stop Bits	1 bit
Flow control	None

Dynamic Baud Rate Change

Tools updating MRAM (updateSystemPackage, app-write-mram) will increase the baud rate from the default value to 921600 baud. Once the operation is completed the Baud rate is set back to the default value.

Use the -s' option to override this dynamic change facility.

SES supports MRAM updates at the standard HF RC/XO frequencies, with baud rate bumping. I.e., no need to specify '-s'. If the device is running at a non-standard frequency, via the 1/X or 1/Z dividers - the MRAM updates should be done via the '-s' switch.

Baud Rate Override

The ISP method uses a default baud rate. For REV_A1 devices this is 100000.

You can override the default baud rate by doing either:

- Edit the local configuration file isp_config_data.cfg and change baudrate.
- Use the -b option e.g. -b 55000

All ISP tools have this option.

UART Errors – PermissionError

```
updateSystemPackage.py
Burning: System Package in MRAM
Selected Device:
Part# E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B4

Connecting to the target device...
[INFO] baud rate 55000
[INFO] dynamic baud rate change Enabled
[ERROR] openSerial could not open port 'COM7': PermissionError(13, 'Access is denied.', None, 5)
[ERROR] isp openSerial failed for COM7
```

There is only one SE-UART on the device. If you are using ISP, please ensure you have no other Tera term or putty sessions using the same SE-UART. The following shows the output if the SE-UART is already being used by another program:

This indicates that the UART is already being used (e.g., a Tera-Term session is still running).

UART Errors – Disconnected

This error occurs when a Disconnect event occurs such as Powering off the FDTI UART whilst in an ISP operation.

```
Available options:
1 - Hard maintenance mode
2 - Soft maintenance mode
3 - Device reset

Select an option (Enter to return): 1

[ERROR] /dev/ttyUSB0 readSerial reporting disconnected
```

In this example, we are in Hard Maintenance mode waiting for input from the Target board. The UART power is turned off and we will see the following error message printed that there has been a disconnecting event.

UART Errors – FileNotFoundError

```
$ ./maintenance
[ERROR] openSerial could not open port 'COM8': FileNotFoundError(2, 'The system cannot find the file specified.', None, 2)
[ERROR] isp openSerial failed for COM8
```

This error usually indicates that your target system is not powered.

ISP Discovery

This command discovers the available Serial communication ports. The first time you execute an SETOOLS script you will be prompted for the required serial port.

When the ports are presented, just enter the port name and press [ENTER]. Just pressing [ENTER] will pick the last item in the list.

This port data is saved in a local configuration file (isp_config_data.cfg).

The next time a SETOOLS command is invoked if this configuration file is present, it will use the parameters from this file.

To override this option simply use the -d option:

```
$ ./app-write-mram -d
Writing MRAM with parameters:
Device Part# E7 (AE722F80F55D5AS) - 5.5 MRAM / 13.5 SRAM - Rev: B2
- Available MRAM: 5767168 bytes
[INFO] Burning: ../build/AppTocPackage.bin 0x8057c4e0
[INFO] baud rate 55000
[INFO] dynamic baud rate change Enabled
COM ports detected = 2
-> COM8
-> COM9
Enter port name:|
```

This will force a re-discovery of the Serial ports.

All the MRAM ISP tools have the discovery option which will prompt you for the serial port.

ISP when in Low power mode

When an application has put the device into STOP power mode, ISP will not be operational (the device is Powered off).

8. Tool Flow

The tools allow a user to provision the device.

This includes programming MRAM with application binaries for A32 and M55 embedded processors available for user applications.

- a. Use `tools-config` to review and modify current tool options.
- b. Generate RoT with `app-gen-rot.py` only once (or anytime new keys are required) – **Not required** for evaluation purposes as this release packet already contains a sample RoT and corresponding keys and certificates, but it can be used if user wants to re-generate the RoT. However, we strongly recommend generating your own RoT if decide to develop a project.
- c. Edit `build/config/app-cfg.json` file (or a json file with other name) with images to include in the desired **AppTocPackage.bin** image and copy the binaries into the `build/` directory. Run `app-gen-toc.py` to generate the package.
- d. Power on the board and connect the Serial interface for the following steps.
 - d.1 (Optional) Use `updateSystemPackage.py` to update latest Alif Secure Enclave firmware.
 - d.2 Use `app-write-mram.py` to upload the `AppTocPackage.bin` image into the MRAM.
 - d.2.1 (Optional) Reset the board to reboot the chip to verify that the SERAM image loads and boots properly.

Debug Stubs

To enable connection with the debugger such as ARM-DS, the target core needs to be in a running state.

This is achieved by running debug stubs on the cores. If the M55_HE Application core does not have an application loaded for it a debug stub will be installed for it. This is done automatically by SES if no Application for the M55_HE is detected. This is not a TOC based debug stub and so does not appear in any Table of contents enquiries. (NOTE: It is important by wiring instructions directly to the TCM memory of the M55_HE and releasing the core)

For the other APP cores, the user must instantiate debug stubs by entering them in an ATOC configuration file.

You can create an ATOC image with 3 SRAM debug stubs for the 3 CPU cores by running

```
app-gen-toc -f build/config/app-cpu-stubs.json
```

Then erase the application part of MRAM by running

```
app-write-mram -e app
```

And then write the image with 3 debug stubs to MRAM by running

```
app-write-mram
```

Details of these programs are described below.

The following is the SE-UART output on boot when the Application MRAM is not programmed.

Name	CPU	Store Addr	Obj Addr	Dest Addr	Boot Addr	Size	Version	Flags	Time <ms>
SERAM0	CM0+	-----	0x00000120	-----	-----	54976	1.0.0	u s	0.00
SERAM1	CM0+	-----	0x00020B20	-----	-----	54976	1.0.0	u s	0.00
DEVICE	CM0+	0x805C1F20	0x805C1520	-----	-----	760	0.5.5	u U	10.41
A32_DBG	A32_0	0x805C2C20	0x805C2220	0x02000000	0x02000000	644	1.0.0	uLUB	9.69
HP_DBG	M55-HP	0x805C38B0	0x805C2EB0	0x50000000	0x50000000	2256	1.0.0	uLUB	9.95
HE_DBG	M55-HE	0x805C4B80	0x805C4180	0x60000000	0x60000000	2256	9.9.9	uLUB	9.97

Legend: <u><C>ompressed, <L>oaded, <U>erified, <s>kippped verification, ooted, <E>ncrypted, <D>eferred

The debug stubs, denoted by the _DBG names, are loaded by SES for each relevant Application core.

MRAM Burners

Two tools used to write MRAM

- updateSystemPackage
- app-write-mram

One will update the System TOC (STOC) and the other will update the Users Application MRAM area.

When using a generated TOC package this will be guaranteed to correctly aligned and padded for writing to MRAM.

In the case of User supplied images the burning tool will warn of any alignment issues such as not being aligned on 16-byte boundaries. The user will need to correct the issue(s) and re-run the burning tool.

Both tools require that SES is running on the Target device. If you are in Recovery mode, these tools will not work as the ISP protocol command sets are different between SEROM and SERAM.

9. Running the Tools

tools-config tool

From the command prompt.

```
SETOOLS OPTIONS CONFIGURATION

*****
Current configuration
- DEVICE Family: Ensemble - Part#: E8 (AE822FA0E5597LS0) - 5.5 MRAM / 9.75 SRAM - Rev: A0
- MRAM BURNER
  Interface: isp
  JTAG Adapter: J-Link
*****

Available options:
1 - Part#
2 - Revision
3 - Interface
4 - JTAG Adapter
5 - Exit (default)

Please enter the number of your option: |
```

\$ tools-config

This script will show the current tool options and allow for their modification.

The DEVICE Family shows the current default device selection. This is important as other tools will use this default device selection such as app-write-mram and app-gen-toc.

Press the menu number to show available options.

```
Available options:
1 - Ensemble (default)
2 - Balletto

Please enter the number of your option: 1

Available options:
1 - E8 (AE822FA0E5597LS0) - 5.5 MRAM / 9.75 SRAM (default)
2 - E8 (AE822FA0E5597BS0) - 5.5 MRAM / 9.75 SRAM
3 - E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM
4 - E7 (AE722F80F55D5AS) - 5.5 MRAM / 13.5 SRAM
5 - E5 (AE512F80F55D5LS) - 5.5 MRAM / 13.5 SRAM
6 - E5 (AE512F80F5582AS) - 5.5 MRAM / 8.25 SRAM
7 - E5 (AE512F80F55D5AS) - 5.5 MRAM / 13.5 SRAM
8 - E5 (AE512F80F5582LS) - 5.5 MRAM / 8.25 SRAM
9 - E4 (AE402FA0E5597BE0) - 5.5 MRAM / 9.75 SRAM
10 - E3 (AE302F80F55D5LE) - 5.5 MRAM / 13.5 SRAM
11 - E3 (AE302F80F5582LE) - 5.5 MRAM / 8.25 SRAM
12 - E3 (AE302F80F55D5AE) - 5.5 MRAM / 13.5 SRAM
13 - E3 (AE302F80F5582AE) - 5.5 MRAM / 8.25 SRAM
14 - E3 (AE302F80C1557LE) - 1.5 MRAM / 5.75 SRAM
15 - E3 (AE302F40C1537LE) - 1.5 MRAM / 3.75 SRAM
16 - E1 (AE101F4071542LH) - 1.5 MRAM / 4.5 SRAM
17 - E1C (AE1C1F4051920PH) - 1.86 MRAM / 2.0 SRAM
18 - E1C (AE1C1F4051920HH) - 1.86 MRAM / 2.0 SRAM
19 - E1C (AE1C1F40319205H) - 1.86 MRAM / 2.0 SRAM
20 - E1C (AE1C1F1041010PH) - 1.0 MRAM / 1.0 SRAM
21 - E1C (AE1C1F1041010HH) - 1.0 MRAM / 1.0 SRAM
22 - E1C (AE1C1F10410105H) - 1.0 MRAM / 1.0 SRAM
23 - E1C (AE1C1F1040505PH) - 0.5 MRAM / 0.5 SRAM
24 - E1C (AE1C1F1040505HH) - 0.5 MRAM / 0.5 SRAM
25 - E1C (AE1C1F10405055H) - 0.5 MRAM / 0.5 SRAM

Please enter the number of your option: |
```

Choose the desired option by pressing the option number (or press Enter key to choose the default one, shown in parentheses).

Once the tool options are configured, the rest of the tools will take these options automatically.

(NOTE: every time a new Part Number or device Revision is changed, it's recommended to re-generate the APP TOC Package using the app-toc.py tool).

Besides the interactive mode explained above (with menus and selections), there are also command line options available.

Tools-config command line mode

This tool can be driven from the interactive menu or via the command line

Tools-config command line mode Set Part Number

To select the Part# (option 1 in the menu)

```
$ tools-config -p <PART#> (use double quotes)
```

Examples:

```
$ tools-config -p "E7 (AE722F80F55D5AE) - 5.5 MRAM / 13.5 SRAM" -r XX
$ tools-config -p "E1 (AE101F4071542UE) - 1.5 MRAM / 4.25 SRAM" -r XX
$ tools-config -p "e7 (AE722F80F55D5AE) - 5.5 MRAM / 13.5 SRAM" -r XX
```

Tools-config command line mode Set SKU

To select a Part, the SKU# can also be used:

```
$ tools-config -k <SKU#>
```

Examples:

```
$ tools-config -k AE722F80F55D5AE -r XX
$ tools-config -k AE101F4071542UE -r XX
```

Tools-config command line mode Set Revision

To select the Revision (option 2 in the menu)

```
$ tools-config -p REVISION
```

Examples:

```
$ tools-config -r B4 -p "XXXXXXXX"
$ tools-config -r A0 -p "XXXXXXXX"
```

Note: Always use -r together with either -p or -k options to fully identify the desired part. Using only one option may lead to an invalid state because not all revisions exist for all parts.

Auto-config

Using a command line option, tools-config can probe your Target device (via the SEUART) and set the default configuration to that device.

The -a or auto config option uses SEUART to *select the device detected during the probe operation*.

Examples:

Initially, tools-config defaults to E8.

```
$ tools-config -a
```

The tool connects to the target and probes it for information about the Target e.g. Part#, Device ID and Revision information.

In this example, a BALLETO (B1) device has been found; this does not match the default device, so it updates the default to the new device discovered.

```
C:\Projects\app-release-SE_FW_1.109.000\app-release>python3 tools-config.py -a
discovering Part# and Revision
[INFO] COM14 open Serial port success
[INFO] baud rate 57600
SETOOLS OPTIONS CONFIGURATION
=====
Probing target for Part# and Revision...
[INFO] Connecting to target...[INFO] Device connected
[INFO] Updating Part# default from E8 (AE822FA0E5597B50) - 5.5 MRAM / 9.75 SRAM to B1 (AB1C1F4M51820PH0) - 1.8 MRAM / 2.0 SRAM
[INFO] Updating Revision default from A0 to A5
*****
Current configuration
- DEVICE Family: Balletto - Part#: B1 (AB1C1F4M51820PH0) - 1.8 MRAM / 2.0 SRAM - Rev: A5
- MRAM BURNER
Interface: isp
JTAG Adapter: J-Link
*****
Available options:
1 - Part#
2 - Revision
3 - Interface
```

Discovery

The -d or discovery option uses SEUART to *discover* the COM ports available.

```
$ tools-config -d
```

Using -d option, the tool-config will run a COM port discovery and present a list with detected ports;

```
C:\Projects\app-release-SE_FW_1.109.000\app-release>python3 tools-config.py -d
discovering Part# and Revision
[INFO] Discover
COM ports detected = 2
-> COM12
-> COM13
Enter port name:
```

Command Port

This command specifies which communication port to use (in case you already know the available COM ports)

```
$ tools-config -c <Comm port>
```

Examples:

```
$ tools-config -c "COM10"
```

Baud Rate

This command specifies which baud rate to use with the communication port if you are using the Discovery option

```
$ tools-config -b <baudrate>
```

Examples:

```
$ tools-config -b 57600
```

Be aware that different devices might have different default baud rates. The probe operation will use the default baud rate of the selected device. As an example, SETOOLS currently defaults to E8 part, so the default rate is set to 57600. Trying to probe an E7 part, with a default baud rate of 55000, might require using the `-b 55000` option.

```
C:\Projects\app-release-SE_FW_1.109.000\app-release>python3 tools-config.py -a -b 55000
discovering Part# and Revision
[INFO] COM12 open Serial port success
[INFO] baud rate 55000
SETOOLS OPTIONS CONFIGURATION
=====
Probing target for Part# and Revision...
[INFO] Connecting to target...[INFO] Device connected
[INFO] Updating Part# default from E8 (AE822FA0E5597LS0) - 5.5 MRAM / 9.75 SRAM to E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM
[INFO] Updating Revision default from A0 to B4
*****
Current configuration
- DEVICE Family: Ensemble - Part#: E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B4
- MRAM BURNER
Interface: isp
JTAG Adapter: J-Link
*****
Available options:
1 - Part#
2 - Revision
3 - Interface
4 - JTAG Adapter
5 - Exit (default)
Please enter the number of your option:
```

Note: there is no need to specify the `-a` option if you are using any of the related options (`-d`, `-c` or `-b`) as specifying any of the latest one, will enable the `-a` option automatically.

app-gen-rot tool

From the command prompt.

```
$ app-gen-rot (-h for help)
```

The Security Tools already have a development key embedded.

You will need to use the -o, --overwrite flag to create a new one.

The tool will prompt the user to enter a password to encrypt the generated keys before saving them in the disk. The password will be saved as 'oem_keys_pass.pwd' and it will be needed for each operation that requires reading the encrypted keys.

```
ls ./app-gen-rot -o
utils/key/hbkl_hash.txt
utils/key/hbkl_zeros.txt
utils/key/kce.txt
utils/key/OEMRoT.pem
utils/key/OEMRoTPublic.pem
utils/key/OEMSBContent.pem
utils/key/OEMSBContentPublic.pem
utils/key/OEMSBKey.pem
utils/key/OEMSBKeyPublic.pem
utils/key/oem_keys_pass.pwd
cert/OEMSBKey1.crt
cert/OEMSBKey2.crt
cert/SBApp-device-config.bin.crt
cert/SBm55_blink_he.bin.crt
cert/SBm55_he_power_test.bin.crt
cert/SBm55_hp_power_test.bin.crt
Please enter a new password for the keys to generate:silly
Generating APP RoT keys (to be used in Key 1 Certificate)
2023-06-28 15:53:37.367 - RSA Key Generator Utility started (Logging to build/logs/key_gen_log.log)
2023-06-28 15:53:37.724 - Private key written to: utils/key/OEMRoT.pem
2023-06-28 15:53:37.726 - Public key written to: utils/key/OEMRoTPublic.pem
2023-06-28 15:53:37.726 - Script completed successfully
Generating APP SB Key keys (to be used in Key 2 Certificate)
2023-06-28 15:53:37.727 - RSA Key Generator Utility started (Logging to build/logs/key_gen_log.log)
2023-06-28 15:53:38.200 - Private key written to: utils/key/OEMSBKey.pem
2023-06-28 15:53:38.212 - Public key written to: utils/key/OEMSBKeyPublic.pem
2023-06-28 15:53:38.213 - Script completed successfully
Generating APP SB Content keys (to be used in Content Certificates)
2023-06-28 15:53:38.214 - RSA Key Generator Utility started (Logging to build/logs/key_gen_log.log)
2023-06-28 15:53:38.446 - Private key written to: utils/key/OEMSBContent.pem
2023-06-28 15:53:38.449 - Public key written to: utils/key/OEMSBContentPublic.pem
2023-06-28 15:53:38.449 - Script completed successfully
Generating APP Hbkl
2023-06-28 15:53:38.451 - Hbkl Generator Utility started (Logging to build/logs/gen_hbkl_log.log)
2023-06-28 15:53:38.451 - Step 1: Calculating hash
2023-06-28 15:53:38.452 - Step 2: Calculate num of zero bits over the hash
2023-06-28 15:53:38.454 - Script completed successfully
Generating APP SB Key 1 Certificate
2023-06-28 15:53:38.467 - Key Certificate Generation Utility started (Logging to build/logs/OEMSBKey1.log)
2023-06-28 15:53:38.468 - Parsing config file: utils/cfg/OEMSBKey1.cfg
2023-06-28 15:53:38.471 - Parsed items:
[('cert-keypair', 'utils/key/OEMRoT.pem'), ('cert-keypair-pwd', 'utils/key/oem_keys_pass.pwd'), ('hbkl-id', '1'), ('nvc-counter-val', '0'), ('next-cert-pubkey', 'utils/key/OEMSBKeyPublic.pem'), ('cert-pkg', 'cert/OEMSBKey1.crt')]
2023-06-28 15:53:38.472 - Raw content of config files:
# This configuration file is for the offline key certificate tool cert_key_util.py (Key Certificate Generation Tool - KCGT).
```

```

2023-06-28 15:53:38,472 - Raw content of config file:
# This configuration file is for the offline key certificate tool cert_key_util.py (Key Certificate Generation Tool - KCGT).
#
# The available parameters in this configuration file are the following
# @@ [KEY-CFG] : Mandatory header.
# The internal non-configurable header.
# @@ cert-keypair : Mandatory parameter.
# The file holding the RSA keypair for signing this certificate, in PEM format.
# @@ cert-keypair-pwd : Optional. If omitted the tool prompts for direct input.
# The passphrase file for the keypair file, in .txt format.
# @@ hbk-id : Mandatory parameter. The tool is agnostic to the certificate usage, this parameter cannot be omitted.
# The ID of the Hbk field in OTP memory that the public key of this certificate is verified against:
# - 0: 128-bit Hbk0.
# - 1: 128-bit Hbk1.
# - 2: 256-bit Hbk.
# The ROM code uses this field only if this certificate is the first certificate in:
# - A two-level SB certificate chain.
# - A three-level SB certificate chain.
# - A three-level Secure Debug chain.
# @@ nvcounter-val : Mandatory parameter.
# The NV counter value:
# - 0..64: the ICV counter.
# - 0..96: the OEM counter.
# - 0..160: the full counter, if OEM and ICV are a single entity.
# The passphrase file for the keypair file, in .txt format.
# @@ next-cert-pubkey : Mandatory parameter.
# The file holding the RSA public key for signing the next certificate in the chain, in PEM format.
# @@ cert-pkg : Mandatory parameter.
# The key certificate package output file.
[KEY-CFG]
cert-keypair = utils/key/OEMRoT.pem
cert-keypair-pwd = utils/key/oem_keys_pass.pwd
hbk-id = 1
nvcounter-val = 0
next-cert-pubkey = utils/key/OEMSBKeyPublic.pem
cert-pkg = cert/OEMSBKey1.crt

2023-06-28 15:53:38,474 - **** Creating Key certificate ****
2023-06-28 15:53:38,482 - Write the certificate to file
2023-06-28 15:53:38,484 - **** Certificate file creation has been completed successfully ****
Generating APP SB Key 2 Certificate
2023-06-28 15:53:38,485 - Key Certificate Generation Utility started (Logging to build/logs/OEMSBKey2.log)
2023-06-28 15:53:38,486 - Parsing config file: utils/cfg/OEMSBKey2.cfg
2023-06-28 15:53:38,493 - Parsed items:
[('cert-keypair', 'utils/key/OEMSBKey.pem'), ('cert-keypair-pwd', 'utils/key/oem_keys_pass.pwd'), ('hbk-id', '1'), ('nvcounter-val', '0'), ('next-cert-pubkey', 'utils/ke
y/OEMSBContentPublic.pem'), ('cert-pkg', 'cert/OEMSBKey2.crt')]
2023-06-28 15:53:38,494 - Raw content of config file:
# This configuration file is for the offline key certificate tool cert_key_util.py (Key Certificate Generation Tool - KCGT).
#
# The available parameters in this configuration file are the following
# @@ [KEY-CFG] : Mandatory header.
# The internal non-configurable header.
# @@ cert-keypair : Mandatory parameter.
# The file holding the RSA keypair for signing this certificate, in PEM format.
# @@ cert-keypair-pwd : Optional. If omitted the tool prompts for direct input.
# The passphrase file for the keypair file, in .txt format.
# @@ hbk-id : Mandatory parameter. The tool is agnostic to the certificate usage, this parameter cannot be omitted.
# The ID of the Hbk field in OTP memory that the public key of this certificate is verified against:
# - 0: 128-bit Hbk0.
# - 1: 128-bit Hbk1.
# - 2: 256-bit Hbk.
# The ROM code uses this field only if this certificate is the first certificate in:
# - A two-level SB certificate chain.
# - A three-level SB certificate chain.
# - A three-level Secure Debug chain.
# @@ nvcounter-val : Mandatory parameter.
# The NV counter value:
# - 0..64: the ICV counter.
# - 0..96: the OEM counter.
# - 0..160: the full counter, if OEM and ICV are a single entity.
# The passphrase file for the keypair file, in .txt format.
# @@ next-cert-pubkey : Mandatory parameter.
# The file holding the RSA public key for signing the next certificate in the chain, in PEM format.
# @@ cert-pkg : Mandatory parameter.
# The key certificate package output file.
[KEY-CFG]
cert-keypair = utils/key/OEMSBKey.pem
cert-keypair-pwd = utils/key/oem_keys_pass.pwd
hbk-id = 1
nvcounter-val = 0
next-cert-pubkey = utils/key/OEMSBContentPublic.pem
cert-pkg = cert/OEMSBKey2.crt

2023-06-28 15:53:38,497 - **** Creating Key certificate ****
2023-06-28 15:53:38,505 - Write the certificate to file
2023-06-28 15:53:38,506 - **** Certificate file creation has been completed successfully ****
Check logs in build/logs/ directory
Done!

```

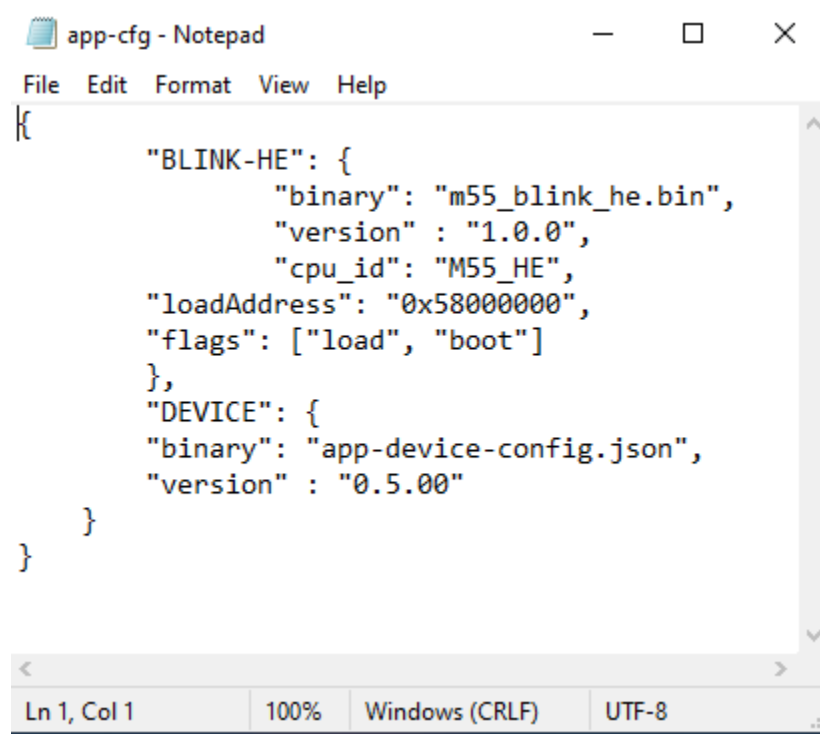
Once finished, keys will be generated in *utils/key/*, certificates are generated in *cert/*, and logs in *build/logs/* folder.

NOTE: Once you generate your RoT, make sure to back it up—especially if you use it to provision devices. If it's lost, you won't be able to create signed packages for those devices. Additionally, whenever you install a new version of SETOOLS, you'll need to restore your RoT to replace the default one included with the tools.

app-gen-toc tool

To generate the APP TOC Package, the `build/config/app-cfg.json` file must be configured. Alternatively, a configuration file can be created with a different name and the tool's `-f` option can cause that alternate file to be used as input to the program.

Here is an example configuration file:



```
{
  "BLINK-HE": {
    "binary": "m55_blink_he.bin",
    "version" : "1.0.0",
    "cpu_id": "M55_HE",
    "loadAddress": "0x58000000",
    "flags": ["load", "boot"]
  },
  "DEVICE": {
    "binary": "app-device-config.json",
    "version" : "0.5.00"
  }
}
```

This configuration file is in JSON format and must specify each of the binary images the user wants to include in the APP TOC Package. [SERAM will process the images in the specified order \(from top to bottom\).](#)

A JSON file may contain multiple objects, where each object is represented by a set of name-value pairs.

In the configuration file, each image must be specified as an object in the following way;

```
"IMAGE_IDENTIFIER": {  
    "attribute_1": "value_1",  
    "attribute_2": "value_2",  
    .....  
    "attribute_n": "value_n"  
}
```

Multiple images will be declared by multiple objects, separated by comma. For example.

```
{  
    "IMAGE_1": {  
  
    },  
    "IMAGE_2": {  
  
    },  
    .....  
    "IMAGE_N": {  
  
    }  
}
```

The "IMAGE_IDENTIFIER" is an 8-character length (MAX) description intended to give a short description to the image. The tool will truncate or extend this field's length as appropriate. This name **should be unique** per configuration file.

Inside each image object, different attributes can be used to define the parameters of the binary image. The following attributes are valid:

binary: [MANDATORY] - this attribute declares the name of the binary to be included in the APP TOC Package. It is assumed the binary exists in the build/images/ folder.

version: [OPTIONAL] – this attribute declares the version of the image, and it should follow the 'X.Y.Z' format. Example: '1.0.0'

loadAddress: [MANDATORY if LOAD flag is set] – this attribute specifies the loading address in (RAM) memory where the code will be executed from. Global Memory Addresses (in hexadecimal format) should be used for this attribute.

mramAddress: [OPTIONAL] – this attribute specifies the MRAM address location where the user wishes the image to reside, and addresses (in hexadecimal format) starting from 0x8000-0100 (for REV_Ax) can be used. This option is to support the "Fixed Address Image" or "absolute placement" use case, where the user needs to position the binary in a specific address in MRAM, so other running code can locate this image in the specified location (Linux use case). If this attribute is used with the BOOT flag, XIP (execution in place) is supported. If this flag is not set, then the image will reside at this address, with no further implication. If this attribute is NOT specified, the tool will define the MRAM position to allocate the binary in memory.

flags: [OPTIONAL] – the following flags can be used – example [“load”, “boot”]:

- LOAD – The image will be loaded into the specified memory address
- BOOT – The specified CPU_ID will be started to execute the image
- ENCRYPT – The image will be encrypted
- COMPRESS – The image will be compressed
- DEFERRED – The image will be skipped at boot time (i.e., no boot or load) and wait for a service request at runtime.

cpu_id: [OPTIONAL] – this attribute indicates the CPU to start once the binary is ready to be executed. Valid values are:

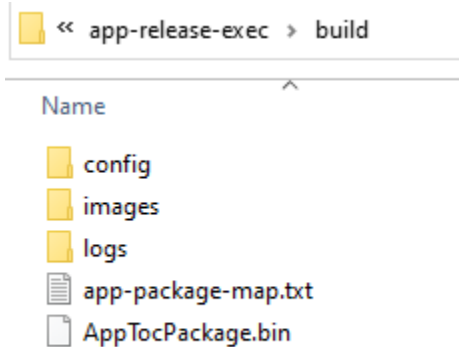
[A32_0, A32_1, A32_2, A32_3, M55_HP, M55_HE]

Note: Only one CPU_ID should be specified.

signed: [OPTIONAL] – this attribute is a binary value (true/false) specifying if the image is signed (“signed”: true) or unsigned (“signed”: false). In a typical case for Secure Boot, all images should be signed, so there is no need to specify this attribute. Only if the user wishes to specify an unsigned image, this attribute should be set as false.

disabled: [OPTIONAL] – this attribute allows a temporary disable of an image in the configuration file for testing purposes, so there is no need to delete the object for that image. Specifying “disabled”: true, the tool will ignore the entry. Either deleting the attribute, or setting it as false, the tool will include back the image in the final APP Package.

The JSON configuration file, along with all the binaries declared in it, should be allocated in the build/ folder (build/config/ and build/images respectively);



Once the *app-cfg.json* file is configured as desired, we need to be sure that all declared images do exist in the *build/images/* folder.

After running the *app-gen-toc.py* tool, the **APP TOC Package** (AppTocPackage.bin) will be generated in the *build/* folder.

The *build/logs/* folder contains a log for the Generation (OEMSBContent.log file)

Once the configuration is done, and all declared images exist in *build/images/* folder, continue with the APP TOC Package generation step.

From the command prompt.

```
$ app-gen-toc (-h for help)
```

```
$ ./app-gen-toc
Generating APP Package with:
Device Part# E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B2
- System MRAM Base Address: 0x80580000
- APP MRAM Base Address: 0x80000000
- APP MRAM Size: 5767168
- Configuration file: build/config/app-cfg.json
- Output file: build/AppTocPackage.bin

Generating Device Configuration for: app-device-config.json
Calculating APP area...
Creating Content Certificates...
2024-01-26 13:46:31,300 - Content Certificate Generation Utility started (Logging to ../build/logs/ICVSBContent.log)
2024-01-26 13:46:31,561 - Content Certificate Generation Utility started (Logging to ../build/logs/ICVSBContent.log)
Creating APP TOC Package...
Adding ATOC...
APP TOC Package size: 15104 bytes
Creating Signature...
Binary File: ../build/AppTocPackage.bin
2024-01-26 13:46:31,809 - Content Certificate Generation Utility started (Logging to ../build/logs/ICVSBContent.log)
Content Certificate File: build/AppTocPackage.bin.crt
Signature File: build/AppTocPackage.bin.sign
Done!
```

Once the tool finishes, a map file named *app-package-map.txt* will be created in build/ folder to provide a reference of the memory map and images/certificates packaged in the APP TOC Package created by the tool. This memory map file will have the following structure.

```
oem-package-map.txt - Notepad
File Edit Format View Help
*** User managed MRAM locations ***
0xa0000100      0x280      640      signed      a32_stub_0.bin

*** OEM Package Start Address ***

- Certificates for User managed images
0xa057d670      0x690      1680      Key1/Key2 Certificates
0xa057dd00      0x370      880      SBa32_stub_0.bin.crt

- Certificates & images for Tool managed images
0xa057e070      0x370      880      SBm55_stub_hp.bin.crt
0xa057e3e0      0x8d0      2256      m55_stub_hp.bin
0xa057ecb0      0x690      1680      Key1/Key2 Certificates
0xa057f340      0x370      880      SBm55_stub_he.bin.crt
0xa057f6b0      0x8d0      2256      m55_stub_he.bin

- OEM TOC (Table of Content)
0xa057ff80      0x10      16      OEM TOC Header
0xa057ff90      0x20      32      OEM TOC entry for A32_0      obj_a
0xa057ffb0      0x20      32      OEM TOC entry for M55_HP      obj_a
0xa057ffd0      0x20      32      OEM TOC entry for M55_HE      obj_a
0xa057fff0      0x10      16      OEM TOC Tail

OEM Package Summary:
- OEM Package total size: 10640 bytes
- OEM Package Start Address: 0xa057d670
- OEM TOC size: 128 bytes
- OEM TOC Start Address: 0xa057ff80
- OEM CRC32: 0xe25282ff

*** END of OEM Package ***
```

Additionally, a script file will be created in bin/ folder to automate the burn of this package into MRAM, using the app-write-mram.py tool. See next section for details.

External OSPI memory support

For those devices that support external OPSI memory booting, the app-gen-toc tool can generate the APP Package to run from this memory using:

```
$ app-gen-toc -m ospiX:SS
```

Where:

- X is OSPI interface (0 or 1)
- SS is the memory size (32, 62, 132, etc.) expressed in MB (megabytes).

An example would be

```
$ app-gen-toc -m ospi0:32
```

- <ospiX> = Use OSPI0 interface
- <SS> = 32MB size

Once the tool is invoked with these options, check the oem-package-map.txt to verify the memory map addresses.

Valid LoadAddress Regions

The valid address ranges supported are as follows:

Start Address	Size	Name	Name
0x5000-0000	256KB	SRAM2	M55_HP_ITCM_BASE
0x5080-0000	1MB	SRAM3	M55_HP_DTCM_BASE
0x5800-0000	256KB	SRAM4	M55_HE_ITCM_BASE
0x5880-0000	256KB	SRAM5	M55_HE_DTCM_BASE
0x6000 0000	256KB	SRAM9_A	
0x6004 0000	1024KB	SRAM6_A	
0x6200 0000	1024KB	SRAM6_B	
0x6300 0000	512KB	SRAM7	
0x6320 0000	2048KB	SRAM8	
0x8000-0000	6MB	MRAM	MRAM_BASE
0xA000-0000	custom	OSPI0	OSPI Ext. Memory Interface 0
0xC000-0000	custom	OSPI1	OSPI Ext. Memory Interface 1

Note on Encryption and XIP

Executable images that are both XIP and Encrypted are not supported as the Crypto hardware doesn't support decryption in NVM. The image should be executed out of TCM if marked as Encrypted.

Note on Factory (Blinky) for ENSEMBLE E1C and BALLETTTO B1

Security toolset (SETOOLS) contains a Factory Image ("Blinky") for ALIF development boards. This is supplied as a binary and is intended as a quick visual test of SETOOLS.

The ENSEMBLE E1C and BALLETTTO B1 series devices differ from the ENSEMBLE E1, E3, E5 and E7 series devices. This means that the default Blinky for an ENSEMBLE device (E1,3,5,7) does not run on an ENSEMBLE E1C device, main difference being the Memory layouts.

For E1C/B1C devices there are added configuration files as configuration files for Balletto devices (both "Ensemble" E1C and B1).

- app-cfg-1c.json
- app-device-config-1c.json

The app-cfg-1c.json file uses m55_blink_he_1c.bin binary, tailored for Balletto Devkit board. Also, the specific device configuration file is being used in this app configuration.

In SETOOLS, ENSEMBLE is the default family/part. So, to burn the Blinky example to a BALLETTTO Dev board, the following steps should be followed:

- Select a Balletto B1C (or Ensemble E1C) part using the tools.config.py tool.
- Build the ATOC with:

```
$./app-gen-toc -f build/config/app-cfg-1c.json
```

- Burn the ATOC using:

```
$./app-write-mram
```



Alif Security Toolkit User Guide

The LED should blink in the BALLETTTO Dev Kit

app-write-mram tool

- This utility uses ISP (In System Programming) via the SE-UART connection to update the MRAM.

From the Windows Command prompt.

```
$ app-write-mram (-h for help)
```

```
$ ./app-write-mram.py -h
usage: app-write-mram.py [-h] [-b BAUDRATE] [-c COMPORT] [-d] [-e ERASE] [-f] [-F] [-i IMAGES]
                        [-a] [-S] [-s] [-p] [-nr] [-V] [-v]

NVM Burner for Application TOC Package

options:
  -h, --help            show this help message and exit
  -b BAUDRATE, --baudrate BAUDRATE
                        serial port baud rate
  -c COMPORT, --comport COMPORT
                        Specify COM port
  -d, --discover        COM port discovery
  -e ERASE, --erase ERASE
                        ERASE [APP | <start address> <size> [<pattern>] ]
  -f, --full_ospi_erase
                        Full Erase OSPI memory
  -F, --full_ospi_erase_no_write
                        Full Erase OSPI memory without writing to MRAM
  -i IMAGES, --images IMAGES
                        images list to burn into NVM ("/path/image1.bin 0x80001000
                        /path/image2.bin 0x80003000")
  -a, --auth_image      authenticate the image by sending its signature file
  -S, --skip            write ATOC only - skip user managed images
  -s, --switch          dynamic baud rate switch toggle, default=on
  -p, --pad             pad the binary if size is not multiple of 16
  -nr, --no_reset       do not reset target before operation
  -V, --version          Display Version Number
  -v, --verbose          verbosity mode
```

Before running this utility, be sure the CPU board is powered on and the SEUART is connected to the PC. Open Windows Command prompt and change directory to the release directory. Call the script as the example shows.

```
$ ./app-write-mram
Writing MRAM with parameters:
Device Part# E7 (AE722F80F55D5AS) - 5.5 MRAM / 13.5 SRAM - Rev: B2
- Available MRAM: 5767168 bytes
[INFO] Burning: ../build/AppTocPackage.bin 0x8057c4e0
[INFO] baud rate 55000
[INFO] dynamic baud rate change Enabled
[INFO] COM8 open Serial port success
Maintenance Mode = Enabled
Authenticate Image: False
../build/AppTocPackage.bin [#####]100%: 15136/15136 bytes
0.32 seconds
```

The first time this command is executed it will prompt the user for the Serial Communications port name. This information is stored in a local file and will be read again when this command is executed.

You can override this using the -d (discovery) option and you will be prompted again for the port name.

There is a verbose mode (-v) which will print all the communications between the host and the target. During this mode, the progress bar is suppressed.

In SE LCS, the SE firmware requires that the ATOC signature (AppTocPackage.bin.sign) is sent as well. This is done by passing the option '-a' (Authenticate Image) to app-write-mram.py.

```
$ app-write-mram -a
```

Maintenance Mode Internals

When updating MRAM (using app-write-mram or updateSystemPackage) SES is placed into a forced Maintenance mode. This means all cores are stopped being achieved by resetting the device and then entering maintenance mode. This ensures no cores are operating when the MRAM update is performed.

Once the update is completed, the whole device is reset (this is done via an ISP command).

Maintenance Mode reporting

This tool will also report the Maintenance state of the Target system. See the section on the Maintenance tool for more details.

```
$ ./app-write-mram
Writing MRAM with parameters:
Device Part# E7 (AE722F80F55D5AS) - 5.5 MRAM / 13.5 SRAM - Rev: B2
- Available MRAM: 5767168 bytes
[INFO] Burning: ../build/AppTocPackage.bin 0x8057c4e0
[INFO] baud rate 55000
[INFO] dynamic baud rate change Enabled
[INFO] COM8 open Serial port success
Maintenance Mode = Enabled
Authenticate Image: False
../build/AppTocPackage.bin [#####]100%: 15136/15136 bytes
0.32 seconds
```

In the above, the maintenance mode is *Enabled*.

Erase option

Erase option will allow, via ISP, the contents of the Application MRAM to be erased. Erasing means that the MRAM location is written with zeros. The writing is verified. See a later description on how to add your own pattern.

Erasing all the application MRAM

```
$ app-write-mram -e app
```

```
$ ./app-write-mram -e app
Writing MRAM with parameters:
Device Part# E7 (AE722F80F55D5AS) - 5.5 MRAM / 13.5 SRAM - Rev: B2
- Available MRAM: 5767168 bytes
[INFO] Erasing: erase 0x80000000 0x580000
[INFO] baud rate 55000
[INFO] dynamic baud rate change Enabled
[INFO] COM8 open Serial port success
[INFO] erase 0x80000000 5767168 (5,767,168)
```

Depending on the size of the Application MRAM this can take a few seconds to complete, do not panic, the tool will exit when the operation is completed, or an error is detected.

NOTE: The pattern written is always zero and cannot be changed.

Erasing a specific address of application MRAM

The user can specify a specific address and length to erase.

Erasing a specific address of application MRAM

The user can specify a specific address and length to erase.

```
$ ./app-write-mram -e "0x80000000 0x10"
Writing MRAM with parameters:
Device Part# E7 (AE722F80F55D5AS) - 5.5 MRAM / 13.5 SRAM - Rev: B2
- Available MRAM: 5767168 bytes
[INFO] Erasing: erase 0x80000000 0x10
[INFO] baud rate 55000
[INFO] dynamic baud rate change Enabled
[INFO] COM8 open Serial port success
[INFO] erase 0x80000000 16 (16)
```

This will erase from the start of MRAM for 16 Bytes, it will be erased with zeros.

Erasing a specific address of application MRAM with a pattern

The user can specify a specific address and length to erase along with a pattern to be written.

```
$ ./app-write-mram -e "0x80000000 0x10 0xa5a5a5a5"
Writing MRAM with parameters:
Device Part# E7 (AE722F80F55D5AS) - 5.5 MRAM / 13.5 SRAM - Rev: B2
- Available MRAM: 5767168 bytes
[INFO] Erasing: erase 0x80000000 0x10 0xa5a5a5a5
[INFO] baud rate 55000
[INFO] dynamic baud rate change Enabled
[INFO] COM8 open Serial port success
[INFO] erase 0x80000000 16 (16)
```

In this example, the pattern 0xa5a5a5a5 will be written to the <address> for <length> bytes.

Erasing a specific address of MRAM that is illegal.

If the user specifies a region of MRAM that is not allowed to be accessed

```
$ ./app-write-mram -e "0x80580000 0x10"
Writing MRAM with parameters:
Device Part# E7 (AE722F80F55D5AS) - 5.5 MRAM / 13.5 SRAM - Rev: B2
- Available MRAM: 5767168 bytes
[INFO] Erasing: erase 0x80580000 0x10
[INFO] baud rate 55000
[INFO] dynamic baud rate change Enabled
[INFO] COM8 open Serial port success
[INFO] erase 0x80580000 16 (16)
[ERROR] illegal address 0x80580010 (0x80580000 + 0x10)
```

In this example, an error is flagged that the address is illegal.

External OSPI Memory Boot support

For those devices that support External OSPI Memory for booting, and they have the appropriate option configured in the OTP memory, then the app-write-mram tool can be used to upload the APP Package to the OSPI memory.

How it works

app-write-mram (without command options) will take the information from build/app-package-map.txt and use those addresses to write in the corresponding memory (MRAM or OSPI). If the

address is in OSPI space, it will first erase OSPI with the size of the image. If image size is smaller than 4KB, a minimum 4KB area will be erased (and starting address will be recalculated to fit this scenario). Erase uses 128K, 32K and 4K (in this order of priorities) to accelerate the erase process.

Using -f option, the tool will perform a FULL erase (APP memory space in OSPI memory).

Using -F option, the tool will perform a FULL erase, without writing the image to the OSPI memory (equivalent to -e APP option in MRAM)

Options -i IMAGES is still valid for OSPI but make sure addresses correspond to OSPI space

Padding binaries (-p option)

The MRAM controller requires write operations with data size in multiple of 16 bytes.

The tools take care of this requirement by always producing images that satisfy this constraint.

In case the user specifies binaries to be placed in specific MRAM addresses (using the “mramAddress” attribute in the **app-gen-toc** configuration file), these binaries should also satisfy the MRAM controller requirement (i.e., the size should be multiple of 16 bytes). When running the **app-write-mram** tool, if a user binary does not satisfy this constraint, the tool will throw an error message and abort the process.

The user should invoke the tool using the -p (or --pad) option to force the tool to pad the binaries. Be aware that using this option, the user provided binary will be modified (padded) and no backup will be created.

No Reset option (-nr)

The Reset option causes the command to reset the target and put it in ‘Maintenance mode’, in which no CPU cores are running (the TOCs are not processed). This is useful in cases where it is not desirable to have them running while writing to MRAM. To overwrite this mode (default), use -nr option (no reset)

```
$ app-write-mram -nr
```

```
$ ./app-write-mram -nr
Writing MRAM with parameters:
Device Part# E7 (AE722F80F55D5A5) - 5.5 MRAM / 13.5 SRAM - Rev: B2
- Available MRAM: 5767168 bytes
[INFO] Burning: ../build/AppTocPackage.bin 0x8057c4e0
[INFO] baud rate 55000
[INFO] dynamic baud rate change Enabled
[INFO] COM8 open Serial port success
Maintenance Mode = Disabled
Authenticate Image: False
build\AppTocPackage.bin [#####]100%: 15136/15136 bytes
0.33 seconds
```

Note: By default, the reset is enabled, the override is ‘-nr’

Images option (-i)

The -i option was implemented to provide a facility to download individual files at specific addresses. Note, this is not needed to download the *AppTocPackage.bin* file as the tool takes care of this.

Initially JTAG was used (via ARM-DS with ULink Pro) to burn images to MRAM. This uses semi-hosting to download the files. Since the MRAM burner runs from the `app-release/bin` directory, the **file should be specified relative to this path**. Subsequently, ISP support for MRAM burning was developed but still assume this filename approach for compatibility.

As an example, a binary file stored in `app-release/build/images/m55_blink.bin`, should be specified with its *relative path* from the `app-release/bin` directory.

```
$ app-write-mram -i "../build/config/images/m55_blink.bin 0x80001000"
```

Note also that double quotes must be provided.

SKIP Images option (-S)

The -S option will write only the ATOC to MRAM. Any managed images to be written will be ignored.

SWITCH option (-s)

The -s option will toggle the dynamic baud rate change when performing bulk transfer operations. By default, dynamic baud rate change is ENABLED.

No reset option (-nr)

The -nr option will not reset the Target being doing the Application MRAM write.

COMPORT (-c)

```
$ app-write-mram.py -c <comm port>
```

Pass in a specific Serial Communication Port name from the command line. The -c option will override the Serial port default.

```
$ python3 app-write-mram.py -c COM6
Writing MRAM with parameters:
Device Part# E8 (AE822FA0E5597LS0) - 5.5 MRAM / 9.75 SRAM - Rev: A0
- Available MRAM: 5,767,168 bytes
[INFO] Burning: ../build/AppTocPackage.bin 0x8057bff0
[INFO] baud rate 57600
[INFO] dynamic baud rate change Enabled
[INFO] port override COM6
[INFO] COM6 open Serial port success
[INFO] Detected Device:
Part# AE822FA0E5597LS0 - Rev: A0
Maintenance Mode = Enabled
Authenticate Image: False
Download Image
build\AppTocPackage.bin [#####]100%: 16400/16400 bytes
Done
1.12 seconds
```

The first execution of any ISP command will force a Discovery mode where the user is asked to select the detected communications port. Once a successful ISP session has occurred the comm port used will be stored in a local file in the app-release directory. Using this option will override this default.

ATOC Policy

The HFRC is not very precise, and it is possible that on some boards it runs at a quite different frequency from its nominal 76.8MHz, this may result in seeing ISP errors (such as Checksum).

The policy was changed and starting with V83 – the policy when there is no ATOC is to assume that the external crystals HFXO and LFXO are present, and to switch the device to using them, including starting the PLL and switching to it. If the device does not have an external crystal then the device will hang as SES tries to switch to a non-present external crystal.

Part Number and/or Part Revision mismatch warnings

The app-write-mram tool, when using ISP protocol, will probe the device to get the Part# and Revision. If these parameters are different than the one configured in the tools (via tools-config tool), a Warning message will be displayed:

```
python3 app-write-mram.py -b 57600
writing MRAM with parameters:
Device Part# E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B4
- Available MRAM: 5,767,168 bytes
[INFO] Burning: ../build/AppTocPackage.bin 0x8057bff0
[INFO] baud rate 57600
[INFO] dynamic baud rate change Enabled
[INFO] COM10 open Serial port success
[INFO] Detected Device:
Part# AE822FA0E5597LS0 - Rev: A0
[ERROR] ***** Configuration mismatch detected!
Expected: E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM
Detected: E8 (AE822FA0E5597LS0) - 5.5 MRAM / 9.75 SRAM
Continue with operation [y/n]?
> |
```

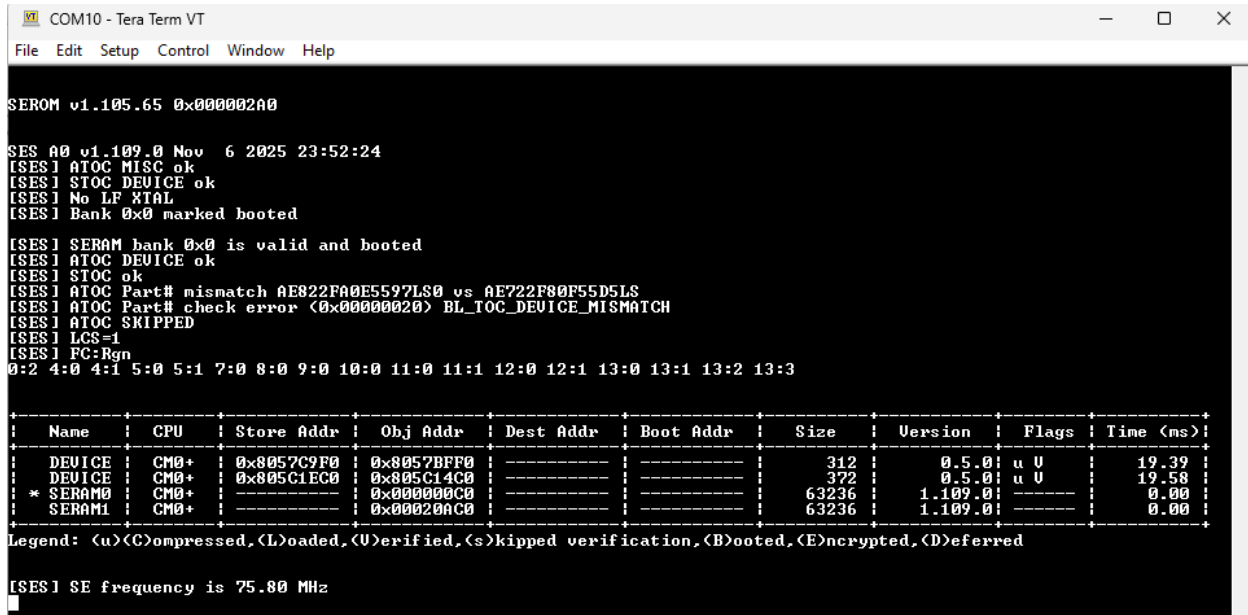
You can press 'Y' to continue (Pressing 'n' or [ENTER] will result in the tool exiting)

```
python3 app-write-mram.py -b 57600
writing MRAM with parameters:
Device Part# E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B4
- Available MRAM: 5,767,168 bytes
[INFO] Burning: ../build/AppTocPackage.bin 0x8057bff0
[INFO] baud rate 57600
[INFO] dynamic baud rate change Enabled
[INFO] COM10 open Serial port success
[INFO] Detected Device:
Part# AE822FA0E5597LS0 - Rev: A0
[ERROR] ***** Configuration mismatch detected!
Expected: E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM
Detected: E8 (AE822FA0E5597LS0) - 5.5 MRAM / 9.75 SRAM
Continue with operation [y/n]?
> y
[ERROR] ***** Part Revision mismatch detected!
Expected: B4
Detected: A0
Continue with operation [y/n]?
> |
```

You can press 'Y' to continue (Pressing 'n' or [ENTER] will result in the tool exiting)

The tool has warned that both the part# and the device revision are not matching but has loaded the binary image to MRAM.

When the device boots up, the SES boot screen will show



```

COM10 - Tera Term VT
File Edit Setup Control Window Help

SEROM v1.105.65 0x000002A0

SES A0 v1.109.0 Nov 6 2025 23:52:24
[SES] ATOC MISC ok
[SES] STOC DEVICE ok
[SES] No LP XTAL
[SES] Bank 0x0 marked booted
[SES] SERAM bank 0x0 is valid and booted
[SES] ATOC DEVICE ok
[SES] STOC ok
[SES] ATOC Part# mismatch AE822FA0E5597LS0 vs AE722F80F55D5LS
[SES] ATOC Part# check error (0x00000020) BL_TOC_DEVICE_MISMATCH
[SES] ATOC SKIPPED
[SES] LCS=1
[SES] FC:Rgn
0:2 4:0 4:1 5:0 5:1 7:0 8:0 9:0 10:0 11:0 11:1 12:0 12:1 13:0 13:1 13:2 13:3

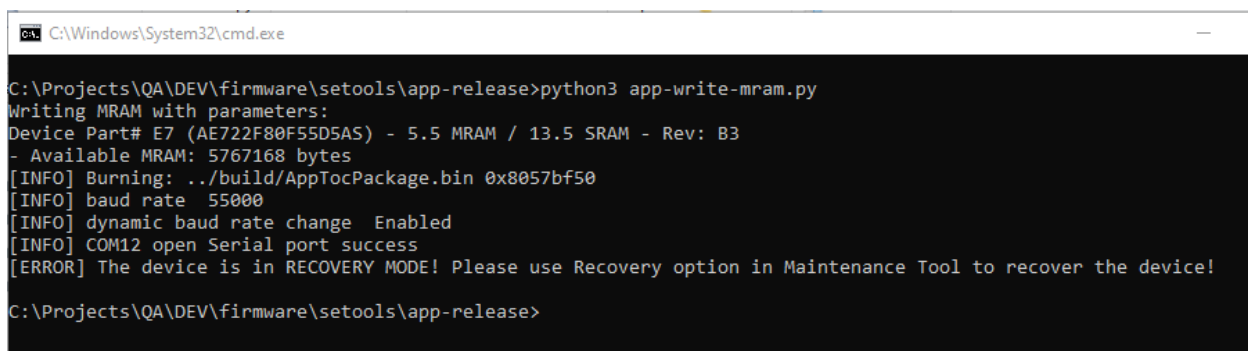
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Name | CPU | Store Addr | Obj Addr | Dest Addr | Boot Addr | Size | Version | Flags | Time (ms)|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| DEVICE | CM0+ | 0x8057C9F0 | 0x8057BFF0 | ----- | ----- | 312 | 0.5.0 | u U | 19.39 |
| DEVICE | CM0+ | 0x805C1EC0 | 0x805C14C0 | ----- | ----- | 372 | 0.5.0 | u U | 19.58 |
| * SERAM0 | CM0+ | ----- | 0x000000C0 | ----- | ----- | 63236 | 1.109.0 | ----- | 0.00 |
| SERAM1 | CM0+ | ----- | 0x00020AC0 | ----- | ----- | 63236 | 1.109.0 | ----- | 0.00 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
Legend: <u><C>ompressed, <L>oaded, <U>erified, <s>kipped verification, <B>ooted, <E>ncrypted, <D>eferred

[SES] SE frequency is 75.80 MHz
  
```

There is now a warning that the ATOC image placed on this device was built for a different device and the part numbers do not match, in this case the ATOC image is not booted or launched and is skipped.

Using app-write-mram in SEROM Recovery mode

If for any reason, the device is in SEROM Recovery mode, the app-write-mram tool will not work as it requires an SES image to be running. The tool will probe the device, via ISP, to check if SEROM or SES is running. If the device is in Recovery mode, then it will warn the user about this condition and exit.



```

C:\Windows\System32\cmd.exe

C:\Projects\QA\DEV\firmware\setools\app-release>python3 app-write-mram.py
Writing MRAM with parameters:
Device Part# E7 (AE722F80F55D5AS) - 5.5 MRAM / 13.5 SRAM - Rev: B3
- Available MRAM: 5767168 bytes
[INFO] Burning: ../build/AppTocPackage.bin 0x8057bf50
[INFO] baud rate 55000
[INFO] dynamic baud rate change Enabled
[INFO] COM12 open Serial port success
[ERROR] The device is in RECOVERY MODE! Please use Recovery option in Maintenance Tool to recover the device!

C:\Projects\QA\DEV\firmware\setools\app-release>
  
```

The user must use the maintenance tool to recover the device.

UpdateSystemPackage tool

Before running this utility, be sure the CPU board is powered on and the SEUART is connected to the PC. Open Windows Command prompt and change directory to the release directory. Call the script as the example shows.

```
$ updateSystemPackage
```

```
Burning: System Package in MRAM
Selected Device:
Part# E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B4

Connecting to the target device...
[INFO] baud rate 55000
[INFO] dynamic baud rate change Enabled
[INFO] COM7 open Serial port success
Bootloader stage: SERAM
[INFO] Detected Device:
Part# AE722F80F55D5LS - Rev: B4
- MRAM Base Address: 0x80580000
Maintenance Mode = Enabled
Authenticate Image: True
Verify Certificate
Signature File: alif\SP-AE722F80F55D5LS-rev-b4-dev.bin.sign
Download Image
alif\SP-AE722F80F55D5LS-rev-b4-dev.bin[#####]100%: 270368/270368 bytes
Verify Image
Done
5.82 seconds

Authenticate Image: True
Verify Certificate
Signature File: alif\offset-58-rev-b4-dev.bin.sign
Download Image
alif\offset-58-rev-b4-dev.bin [#####]100%: 16/16 bytes
Verify Image
Done
0.02 seconds
```

ALIF images are signed. SES will check that image is signed (based on the LCS state of the device) and only allow signed images.

The tool will Probe the target device to ensure it matches the device chosen through tools-config. In the above screen shot, we see:

```
Bootloader stage: SERAM
```

This shows that SERAM is running, the alternative is that SEROM is running and in this case, this is the wrong tool to be using.

```
Device Revision: B4
```

This shows the target device that is attached. The tool will exit if there is a mismatch.

No Reset option (-nr)

The Reset option causes the command to reset the target and put it in 'Maintenance mode', in which no CPU cores are running (the TOCs are not processed). This is useful in cases where it is not desirable to have them running while writing to MRAM. To overwrite this mode (default), use -nr option (no reset)


```
$ updateSystemPackage -nr
```

No Authentication (-na) option

If the device is in DM (Device Manufacturer) or SE (Secure Enable) lifecycle state, images written to the System partition of MRAM must be signed by Alif. The APP release package already contains signatures for the STOC images. The script `updateSystemPackage.py` runs by default in 'Authenticated Mode', i.e., it sends the image signatures to the device for verification. The option 'no-authentication' can be used to override that behavior and run in 'non authenticated mode' instead. This is allowed in CM LCS but not allowed in DM and SE LCSs.

```
$ updateSystemPackage -na
```

Memory option (-m)

The Memory option allows an update to occur to an external Flash memory.

```
$ updateSystemPackage -m
```

The Memory option is only available to devices (e.g. ENSEMBLE E4, E6, E8) that supports External memory Alternate Boot.

```
Burning: System Package in MRAM
Selected Device:
Part# E8 (AE822FA0E5597LS0) - 5.5 MRAM / 9.75 SRAM - Rev: A0

Connecting to the target device...
[INFO] baud rate 57600
[INFO] dynamic baud rate change Enabled
[INFO] COM11 open Serial port success
Bootloader stage: SERAM
[INFO] Detected Device:
Part# AE822FA0E5597LS0 - Rev: A0
[INFO] SERAM in external OSPI was requested
[INFO] Device supports OSPI external memory
[INFO] OSPI is enabled in OTP
ALIF_BASE_ADDRESS 0xc1f80000 alif_offset 0xc1ffff0
- Package: alif\SP-AE822FA0E5597LS0-ospil-32-rev-a0-dev.bin
- Offset: alif\offset-58-ospil-32-rev-a0-dev.bin
OSPI Erase 4.0k sectors [#####]100%: 128/128 sectors
Maintenance Mode = Enabled
Authenticate Image: True
Verify Certificate
Signature File: alif\SP-AE822FA0E5597LS0-ospil-32-rev-a0-dev.bin.sign
Download Image
alif\SP-AE822FA0E5597LS0-ospil-32-rev-a0-dev.bin[#####]100%: 270400/270400 bytes
Verify Image
Done
34.88 seconds

Authenticate Image: True
Verify Certificate
Signature File: alif\offset-58-ospil-32-rev-a0-dev.bin.sign
Download Image
alif\offset-58-ospil-32-rev-a0-dev.bin[#####]100%: 16/16 bytes
Verify Image
Done
0.05 seconds
```

The device must have OTP settings for OSPI for this option to operate.

If the OTP is not programmed, you will see the following

```
Burning: System Package in MRAM
Selected Device:
Part# E8 (AE822FA0E5597LS0) - 5.5 MRAM / 9.75 SRAM - Rev: A0

Connecting to the target device...
[INFO] baud rate 57600
[INFO] dynamic baud rate change Enabled
[INFO] COM11 open Serial port success
Bootloader stage: SERAM
[INFO] Detected Device:
Part# AE822FA0E5597LS0 - Rev: A0
[INFO] SERAM in external OSPI was requested
[INFO] Device supports OSPI external memory
[INFO] OSPI is NOT enabled in OTP

[ERROR] OSPI is not enabled in OTP
```

Setting the target device as the default

You may see the following:

```
Burning: System Package in MRAM
Selected Device:
Part# E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B2

Connecting to the target device...
[INFO] baud rate 55000
[INFO] dynamic baud rate change Enabled
[INFO] COM8 open Serial port success
Bootloader stage: SERAM
[INFO] Detected Device:
Part# AE722F80F55D5LS - Rev: B4
- MRAM Base Address: 0x80580000
Connected target is not the default Revision
Do you want to set this part as default? (y/n): |
```

This is informing you that the target device does not match the default device selected via the tools-config tool.

Answering 'y'es will tell updateSystemPackage to select the correct package for the *detected* device.

Answering 'n'o will tell updateSystemPackage to select the correct package based on the default selection from tools-config.

You can also use tools-config to also set the global device default, it also has dynamic

Maintenance tool

Maintenance mode allows for the recovery of MRAM executable images and interaction with SES.

Running the Tool

```
$ maintenance (-h for help)
usage: maintenance.py [-h] [-b BAUDRATE] [-c COMPORT] [-d] [-opt OPTION] [-V] [-v] ...

SETTOOLS Maintenance Tool

positional arguments:
  params                additional parameters for the command

options:
  -h, --help            show this help message and exit
  -b BAUDRATE, --baudrate BAUDRATE
                        serial port baud rate
  -c COMPORT, --comport COMPORT
                        Specify COM port
  -d, --discover        COM port discovery
  -opt OPTION, --option OPTION
                        -opt <command> [<command>...] | -opt <command> <hex-value> | -opt help
  -V, --version         Display Version Number
  -v, --verbose         verbosity mode
```

The command presents several options including maintenance mode.

Select an option number from the menu.

To exit the tool just press [ENTER] at the option selection.

Command line options (-opt)

This allows a command to be executed by the Maintenance tool without navigating the menu options.

Opt command line format

```

<opt-arg>          ::= "-opt" <simple-command-list>
                    | "-opt" <param-command> <hex-value>
                    | "-opt" "help"

<simple-command-list> ::= <simple-command>
                        | <simple-command> <simple-command-list>

<simple-command>    ::= "getbanner" | "gettoc" | "getrevision" |
                        "getpower"
                        | "getcloc" | "getcpustatus" | "getfirewall"
                        | "geteckey" | "devenquiry"

<param-command>    ::= "getmramdata" | "getotpread"

<hex-value>        ::= "0x" <hex-digits> | <hex-digits>

<hex-digits>       ::= <hex-digit> | <hex-digit> <hex-digits>

<hex-digit>        ::= "0" | "1" | "2" | "3" | "4" | "5" | "6" | "7"
                        | "8" | "9" | "a" | "b" | "c" | "d" | "e" | "f"
                        | "A" | "B" | "C" | "D" | "E" | "F"

$ maintenance -opt <command option> [<offset>]

```

Opt command supported commands

The following commands are supported from the command line

Command	Purpose
getbanner	Returns the SES Banner string
gettoc	Returns the TOC information
getrevision	Returns the device revision
getpower	Returns POWER information
getcloc	Return CLCOK information
getcpustatus	Returns CPU status
getfirewall	Returns Firewall settings
geteckey	Returns ECC key
getmramdata <offset>	Returns MRAM contents at <offset>
getotpread <offset>	Returns OTP contents at <offset>
devenquiry	Returns device state
Help	Show all supported commands

Opt command help

```
$ maintenance.py -opt help
[INFO] COM10 open Serial port success
[INFO] baud rate 57600
[INFO] Connecting to target...Device connected
Supported commands:
  getbanner      Get SES banner
  gettoc         Get TOC info
  getrevision    Get revision info
  getpower       Get power data
  getclock       Get clock data
  getcpustatus   Get CPU boot info
  getfirewall    Get firewall configuration
  getecckey      Get ECC key
  devenquiry     Device enquiry
  getmramdata    Get MRAM <offset>
  getotpread     Get OTP  <offset>
```

Opt command with parameter

For passing an offset

```
$ maintenance.py -opt getmramdata 0x57bff0
```

Opt with multiple commands

```
$ maintenance -opt [<command option> <command option>] [<offset>]
```

```
$ python3 maintenance.py -opt devenquiry getcpustatus gettoc
```

```
Device connected
SERAM Error = 0x0 Extended Error = 0x0 Maintenance Mode = Disabled
```

CPU	Booted	Boot Addr
A32_0		-----
A32_1		-----
M55-HP		-----
M55-HE	YES	0x58000000

Name	CPU	Store Addr	Obj Addr	Dest Addr	Boot Addr	Size	Version	Flags	Time (ms)
DEVICE	CM0+	0x80566b20	0x80566120	-----	-----	312	0.5.0	u v	14.92
DEVICE	CM0+	0x805d1ec0	0x805d14c0	-----	-----	372	0.5.0	u v	15.01
* SERAM0	CM0+	-----	0x000000c0	-----	-----	138836	1.110.0	u s	0.00
SERAM1	CM0+	-----	0x00028ac0	-----	-----	138836	1.110.0	-----	0.00
SRV-HE-T	M55-HE	0x80567660	0x80566c60	0x58000000	0x58000000	100656	1.0.0	uLVB	27.76

For passing an offset

```
$ maintenance.py -opt getcpustatus gettoc getmramdata 0x57bfff0
```

```
Device connected
```

CPU	Booted	Boot Addr
A32_0		-----
A32_1		-----
M55-HP		-----
M55-HE	YES	0x58000000

Name	CPU	Store Addr	Obj Addr	Dest Addr	Boot Addr	Size	Version	Flags	Time (ms)
DEVICE	CM0+	0x80566b20	0x80566120	-----	-----	312	0.5.0	u v	14.92
DEVICE	CM0+	0x805d1ec0	0x805d14c0	-----	-----	372	0.5.0	u v	15.01
* SERAM0	CM0+	-----	0x000000c0	-----	-----	138836	1.110.0	u s	0.00
SERAM1	CM0+	-----	0x00028ac0	-----	-----	138836	1.110.0	-----	0.00
SRV-HE-T	M55-HE	0x80567660	0x80566c60	0x58000000	0x58000000	100656	1.0.0	uLVB	27.76

```
[0x57bfff0] 0x202a2a20 0x2053434c 0x20202020 0x58252020
```

BAUD option (-b)

```
$ maintenance -b <board rate>
```

Pass in a specific Serial Communication Port baud rate from the command line

```
$ maintenance -b 55000
```


COMPORT option (-c)

```
$ maintenance -c <comm port>
```

Pass in a specific Serial Communication Port name from the command line

```
$ maintenance -c COM10
[INFO] port override COM10
[INFO] COM10 open Serial port success
[INFO] baud rate 57600
[INFO] Connecting to target...Device connected
```

Here COM Port “COM10” has been passed. A message states there is a “port override”.

The COM Port details are stored in a file (isp_config_data.cfg) in the SETOOLS directory. After exiting the maintenance tool this file will be updated with, in this case, COM10.

Passing a -d' (Discovery) option will take precedence over the COMPORT command.

Maintenance Menu Grouping

The maintenance menus are grouped into categories.

Action	Menu Groups							
	Device Control	Device Info	MRAM	Utilities	Set	ROM	SEROM	SES
Hard maintenance mode	•							•
Soft maintenance mode	•							•
Device Reset	•							•
Device enquiry		•					•	•
Get revision info		•					•	•
Get TOC info		•						•
Get SES banner		•						•
Get CPU boot info		•						•
Get OTP Data		•						•
Get MRAM data		•					•	•
Get log data		•						•
Get SEROM trace data		•					•	•
Get SERAM trace data		•						•
Get Power data		•						•
Get Clock data		•						•
Get SERAM metrics				•				•
Terminal mode				•			•	•
Get ECC Key				•				•
Get Firewall configuration				•				•
Alternate Boot OTP				•				•
Erase application MRAM			•					•
Fast erase application MRAM			•					•
Fast erase application NTOC			•					
Get MRAM info			•					•
Recovery							•	
Enable LOGGING					•			•
Disable LOGGING					•			•

Enable PRINTING					•			•
Disable PRINTING					•			•

NOTE: Some ISP commands are common between SEROM and SES. When in Recovery mode the number of commands is limited. This is because the SEROM only supports a limited subset of features compared to ISP running from SES.

[maintenance menu in SEROM Recovery mode](#)

If for any reason, the device enters SEROM Recovery mode (i.e. no SES running), the maintenance tool will detect this stage, and it will adapt the menu options to only show those enabled for this case Device Control

Top level group menu

```
Device connected in Recovery
Available options:
1 - ROM
2 - Device Information
3 - Utilities
4 - Alt. Boot Octal SPI
Select an option (Enter to exit):
```

See later section on ROM for details on these options.

[Hard maintenance Mode](#)

Hard maintenance mode will wait until the user has pressed RESET on the target board.

Once [RESET] has been pressed the tool will drop back to the Available options menu. NOTE: This mode skips the TOC processing section of SES, if you use the get TOC info option it will return no data as no TOCs have been loaded. The DEVICE TOCs are processed however,

```
select an option (Enter to return): 1
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Name | CPU | Store Addr | Obj Addr | Dest Addr | Boot Addr | Size | Version | Flags | Time (ms)|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| DEVICE | CM0+ | 0x801c99c0 | 0x801c8fc0 | ----- | ----- | 168 | 0.5.0 | u v | 14.90 |
| DEVICE | CM0+ | 0x801ee850 | 0x801ee4e0 | ----- | ----- | 372 | 0.5.0 | u s | 0.23 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+

```

[Soft maintenance Mode](#)

Soft maintenance mode is like Hard maintenance mode, but no [RESET] button needs to be pressed on the target board.

[Device Reset](#)

Issues a Reset ISP command to the board.

Device Information

Top level group menu

```
Available options:

1 - Get TOC info
2 - Get SES Banner
3 - Get CPU boot info
4 - Device enquiry
5 - Get revision info
6 - Get OTP data
7 - Get MRAM data
8 - Get log data
9 - Get SEROM trace data
10 - Get SERAM trace data
11 - Get power data
12 - Get clock data
```

Get Table of contents information

This will return the TOC contents in MRAM along with the boot status of the Application CPUs.

```
Select an option (Enter to return): 1
```

Name	CPU	Store Addr	Obj Addr	Dest Addr	Boot Addr	Size	Version	Flags	Time (ms)
DEVICE	CM0+	0x805c1ee0	0x805C14E0	-----	-----	404	0.5.0	u V	14.58
SERAM0	CM0+	-----	0x000000E0	-----	-----	78208	1.0.0	-----	0.00
SERAM1	CM0+	-----	0x00020AE0	-----	-----	78208	1.0.0	-----	0.00
HE_DBG	M55-HE	0x805c2a80	0x805C2080	0x58000000	0x58000000	2256	1.0.0	uLVB	14.73

```
Available options:
```

Legend Reference

Name	Name of entry from JSON file.
CPU	Which CPU is this TOC referencing.
Store Address	MRAM location
Object address	MRAM TOC address
Destination address	Copy address for object (if relevant).
Boot Address	Boot address
Size	Size of TOC entry
Version	Version id from JSON file
Flags	Directives for the TOC object
	u Uncompressed image
	C Compressed Image
	L Image is loaded from MRAM
	V Image is verified
	s Image is Skipped
	B Image has been Booted
	E Image is Encrypted
	D Image is Deferred, will be Loaded / Booted later.

When you are in hard maintenance mode, the TOC table contents will look very different. The main reason is that Hard Maintenance skips the TOC processing stage entirely except for some DEVICE CONFIG objects. No Application CPU cores are booted or loaded with images whilst in this mode.

Get SES Banner

```
Select an option (Enter to return): 2
SES A1 v1.111.0 Mar 12 2026 15:45:02
```

This returns the version string for the SES image

NOTE: The Banner output is only 80 characters.

Get CPU boot information

```
Select an option (Enter to return): 3
+-----+-----+-----+
| CPU   |Booted| Boot Addr |
+-----+-----+-----+
| A32_0 |      | ----- |
| A32_1 |      | ----- |
| M55-HP|      | ----- |
| M55-HE| YES  | 0x58000000|
+-----+-----+-----+
```

This returns the status of the CPU cores.

Device Enquiry

Returns if connected to SEROM or SES. Any errors are reported otherwise 0x0 (Success) is returned.

Device Enquiry - SES

```
Select an option (Enter to return): 4
SERAM Error = 0x0 Extended Error = 0x0 Maintenance Mode = Disabled
Available options:
```

It will also tell you if you are in Maintenance mode, in above case it is DISABLED

```
Select an option (Enter to return): 4
SERAM Error = 0x0 Extended Error = 0x0 Maintenance Mode = Enabled
Available options:
```

In the above case it is ENABLED

Device Enquiry – SEROM

When connected to SEROM:

```
Available options:
1 - Get TOC info
2 - Get SES Banner
3 - Get CPU boot info
4 - Device enquiry
5 - Get revision info
6 - Get OTP data
7 - Get MRAM data
8 - Get Log data
9 - Get SEROM trace data
10 - Get SERAM trace data
11 - Get power data
12 - Get clock data

Select an option (Enter to return): 4
SEROM Error = 0x12 (SEROM_ATOC_HEADER_STRING_INVALID) Extended Error = 0x0 Maintenance Mode = <None>
```

NOTE: If you are connected to SEROM then this means your MRAM is not provisioned or an error has occurred booting an SES image. Maintenance mode is not applicable in this scenario.

Get Revision Information

Device Information

Item	Meaning
Version	SOC id register
ALIF_PN	ALIF Part Number
HBK0	SHA-128 of ALIF RoT Public key
HBK1	SHA-128 of Customer RoT Public key
HBK_FW	
Wounding	Shows DISABLED Wounding bits
DCU	Debug Lock bits
MfgData	ALIF Manufacturing data
LCS	Life Cycle State CM, DM, SE, RMA
SES Alt Load	Alternative External memory Load path for SES
SES Ext Conf	Details of the External memory device

NOTE: SES Alt Load and Ext Conf will only appear if you are using ENSEMBLE E4, E6, and E8 devices.

ENSEMBLE E4, E6, E8 Revision A1

Returns device specific information such as the SOC ID, Part# and Alternative External memory boot options.

```
Source      = SERAM
Version     = 0x02A1
ALIF_PN     = 0x0
HBK0        = 00000000000000000000000000000000
HBK1        = 00000000000000000000000000000000
HBK_FW      = 00000000000000000000000000000000
wounding    = 0x0
DCU         = 0x0
MfgData     = 0000000000000000000000000000000000000000000000000000000000000000
LCS         = 0x0 (CM)
SES Alt Load = 00000200
    Loading options : 0x2 External Memory only
    Ext Memory type : 0x0 OSPI Flash 8bit in XIP Mode

SES Ext Conf = 000B0B0B0105E22B00DE2302000000
    rx_sample_delay : 0x0
    xip_incr_inst   : 0xB
    xip_wrap_inst   : 0xB
    aes_rxdelay     : 0xB
    ddr_drive_edge  : 0x1
    pinmux_sel      : 0x5
    bus_speed_sel   : 0x0
    fast_read_wait_cycles: 0x8
    flash_size      : 0x0      32 MB
    valid           : 0x1
    reset_port      : 0xF
    reset_pin       : 0x5
    spi_mode        : 0x0
    io_mode         : 0x0
    ddr_en          : 0x0
    xip_ddr_en      : 0x0
    inst_ddr_en     : 0x0
    xip_inst_ddr_en : 0x0
    rxds_en         : 0x0
    rxds_vl_en      : 0x0
    dfs             : 0xF
    xip_dfs         : 0xB
    xip_rxds_en     : 0x0
    xip_instr_len   : 0x2
    xip_addr_len    : 0x8
    xip_spi_mode    : 0x0
    xip_hyperbus_en : 0x0
    tx_wait_count   : 0x0
```

ENSEMBLE E4, E6, E8 Revision A0

Returns device specific information such as the SOC ID, Part# and Alternative External memory boot options.

In the example below, these settings are default, so the primary boot path is MRAM only.

```
Select an option (Enter to return): 5
Source          = SERAM
Version         = 0x02A0
ALIF_PN         = AE822FA0E5597LS0
HBK0            = CD74AC7009072C3E27B8B8F73AF34EDE
HBK1            = 00000000000000000000000000000000
HBK_Fw         = 00000000000000000000000000000000
wounding        = 0x1
DFT            = Disable
DCU             = 0x0
MfgData         = 0511010C0000020E000001260000000000014806A78A7A40000008800000068
** offset 0x154
+ Valid Data    [31:31] = 1    (0x1)
+ BOR Abort     [30:30] = 0    (0x0)
+ Temp Sensor   [29:24] = 0    (0x00)
+ ADC VREF      [23:16] = 20   (0x14)
+ ADC24 Offset  [15:0 ] = 0    (0x0000)
** offset 0x158
+ AON LDO       [31:28] = 10   (0xA)
+ RET LDO       [27:24] = 4    (0x4)
+ PMU BG        [23:20] = 10   (0xA)
+ PERIPH BG     [19:16] = 7    (0x7)
+ AON BG<4:1>   [15:12] = 7    (0x7)
+ AON BG<0>     [11:11] = 1    (0x1)
+ AON BG (full) [15:11] = 15   (0x0F)
+ Reserved      [10: 8] = 0    (0x0)
+ DCDC Pre-trim [ 7: 0] = 106  (0x6A)
** offset 0x15C
+ HFRC 76.8MHz  [31:26] = 34   (0x22)
+ ADC120 Offset [25:16] = 0    (0x000)
+ LFRC 32.7kHz  [15:10] = 0    (0x00)
+ ADC121 offset [ 9: 0] = 0    (0x000)
** offset 0x160
+ DCDC Post-trim [31:26] = 26   (0x1A)
+ ADC122 Offset [25:16] = 0    (0x000)
+ DAC1 Offset   [15: 8] = 0    (0x00)
+ DAC2 Offset   [ 7: 0] = 0    (0x00)

+ x-loc        12
+ y-loc        1
+ Fab          UAS Global
+ Wafer ID     17
+ Year#       2025
+ Week#       14
+ Lot#        2
LCS            = 0x1 (DM)
SES Alt Load   = 00000000
Loading options : 0x0 MRAM only
SES Ext Conf   = 00000000000000000000000000000000
```

ENSEMBLE Revision B4

Returns device specific information such as the SOC ID, Part number.

```
Select an option (Enter to return): 5
Source           = SERAM
Version          = 0xB400
ALIF_PN          = AE722F80F55D5LS
HBK0             = f049442ecc7900a838d34c31d71e62e6
HBK1             = 00000000000000000000000000000000
HBK_FW           = 00000000000000000000000000000000
Wounding         = 0xc1
    DFT          Disable
    Modem        Disable
    GNSS         Disable
DCU              = 0x0
MfgData          = 0401041500000017000000000000000017809f30d883003c00a0000000088
    + x-loc      21
    + y-loc      4
    + Fab        UAS Global
    + Wafer ID   1
    + Year#      2024
    + Week#      23
    + Lot#       0
LCS              = 0x1 (DM)
```

Analogue Trim decoding

If the device contains Analogue Trimming information, this will be decoded

```

select an option (Enter to return): 5
Source      = SERAM
Version     = 0xA001
ALIF_PN     = AE1C1F4051920PH
HBK0       = AF9771E6F171EE6DE39106BD812BABF0
HBK1       = 00000000000000000000000000000000
HBK_Fw     = 00000000000000000000000000000000000000
wounding    = 0x9
  DFT      Disable
  BLE      Disable
DCU         = 0x0
MfgData     = 0401000100000011000000000000000000001680BC00A8830004009400000078
  ** offset 0x154
    + Valid Data      [31:31] = 1      (0x1)
    + BOR Abort       [30:30] = 0      (0x0)
    + Temp Sensor     [29:24] = 0      (0x00)
    + ADC VREF        [23:16] = 22     (0x16)
    + ADC24 offset    [15:0 ] = 0      (0x0000)
  ** offset 0x158
    + AON LDO         [31:28] = 8      (0x8)
    + RET LDO         [27:24] = 3      (0x3)
    + PMU BG          [23:20] = 10     (0xA)
    + PERIPH BG       [19:16] = 8      (0x8)
    + AON BG<4:1>     [15:12] = 0      (0x0)
    + AON BG<0>       [11:11] = 0      (0x0)
    + AON BG (full)   [15:11] = 0      (0x00)
    + Reserved        [10: 8] = 0      (0x0)
    + DCDC Pre-trim   [ 7: 0] = 188    (0xBC)
  ** offset 0x15C
    + HFRC 76.8MHz    [31:26] = 37     (0x25)
    + ADC120 offset   [25:16] = 0      (0x000)
    + LFRC 32.7kHz    [15:10] = 1      (0x01)
    + ADC121 offset   [ 9: 0] = 0      (0x000)
  ** offset 0x160
    + DCDC Post-trim  [31:26] = 30     (0x1E)
    + ADC122 offset   [25:16] = 0      (0x000)
    + DAC1 offset     [15: 8] = 0      (0x00)
    + DAC2 offset     [ 7: 0] = 0      (0x00)

+ x-loc      1
+ y-loc      0
+ Fab        UAS Global
+ wafer ID   1
+ Year#      2024
+ Week#      17
+ Lot#       0
LCS          = 0x1 (DM)
  
```

Analogue Trim decoding – BALLETO B1 / ENSEMBLE E1C

If the device is BALETO B1 / ENSEMBLE E1C Revision A8 you will see some extra Analogue Trim values ("TON") which are unique to this series for devices.

```
** Offset 0x160
+ DCDC Post-trim [31:26] = 26 (0x1A)
+ TON control    [25:24] = 0  (0x0)
+ TON calib     [23:20] = 0  (0x0)
+ DAC1 Offset   [19:12] = 0  (0x00)
```

Device ID

The EUI48 and EUI64 unique identifiers are shown. This data is derived from the Manufacturing data which is unique for each device.

[illegible]

OTP Integrity

The following fields are checked for integrity

Checked	Integrity
Part Number	Valid ALIF part number
HBK0	Valid ALIF Key
HBK_FW	All zeros
DCU	All zeros

If there is an (unlikely) issue a warning print is issued.

```
Select an option (Enter to return): 2
Source      =  SEROM
Version     =  0xB400
ALIF_PN     =  AE722F88F55L5LS
HBK0        =  4CA48F28D3267E1E520706995F1FA338
HBK1        =  00000000000000000000000000000008
HBK_FW      =  0000000800000000000000000000000000
Wounding    =  0xc1
  DFT       =  Disable
  Modem     =  Disable
  GNSS      =  Disable
DCU         =  0x8000000000000000
MfgData     =  04070C580000001F000001220000000000015808F70B88B003C00A0000000C8
  ** offset 0x154
    + Valid Data    [31:31] = 1      (0x1)
    + BOR Abort     [30:30] = 0      (0x0)
    + Temp Sensor   [29:24] = 0      (0x00)
    + ADC VREF      [23:16] = 21     (0x15)
    + ADC24 offset  [15:0 ] = 0      (0x0000)
  ** offset 0x158
    + AON LDO       [31:28] = 8      (0x8)
    + RET LDO       [27:24] = 11     (0xB)
    + PMU BG        [23:20] = 11     (0xB)
    + PERIPH BG     [19:16] = 8      (0x8)
    + AON BG<4:1>   [15:12] = 7      (0x7)
    + AON BG<0>     [11:11] = 0      (0x0)
    + AON BG (full) [15:11] = 14     (0x0E)
    + Reserved      [10: 8] = 0      (0x0)
    + DCDC Pre-trim [ 7: 0] = 143    (0x8F)
  ** offset 0x15C
    + HFRC 76.8MHz  [31:26] = 40     (0x28)
    + ADC120 offset [25:16] = 0      (0x000)
    + LFRC 32.7kHz  [15:10] = 15     (0x0F)
    + ADC121 offset [ 9: 0] = 0      (0x000)
  ** offset 0x160
    + DCDC Post-trim [31:26] = 50     (0x32)
    + ADC122 offset [25:16] = 0      (0x000)
    + DAC1 offset   [15: 8] = 0      (0x00)
    + DAC2 offset   [ 7: 0] = 0      (0x00)

  + x-loc      88
  + y-loc      12
  + Fab        UAS global
  + Wafer ID   7
  + Year#      2024
  + Week#      31
  + Lot#       0
SerialN      =  0x8000000000000000
LCS          =  0x1 (DM)
Part#        AE722F88F55L5LS      NOT OK
HBK1         00000000000000000000000000000008      NOT OK
HBK_FW       0000000800000000000000000000000000      NOT OK
DCU          00000000000000080000000000000000      NOT OK
```

Get OTP data

Returns OTP data based on a provided OTP word offset

```
Select an option (Enter to return): 6
Enter word addr(hex): 0x70
0x0070 0x0
Available options:
```

In this example, we are reading the first word (0x70) of the OTP area that is available for customer use.

The customer OTP area is between word offsets **0x70 - 0x7F**. If an offset outside that area is specified, an error message is returned:

```
Select an option (Enter to return): 6
Enter word addr(hex): 0
RX<-- length= 4 command= COMMAND_NAK          checksum= 0x10 error= ISP_ERROR_ILLEGAL_ADDRESS
[ERROR] 0x0000 ISP_ERROR_ILLEGAL_ADDRESS
Available options:
```

An error message is reported that you have supplied an ILLEGAL address offset.

Access to the customer OTP area depends on the device's LCS – DM or SE.

OTP area	DM LCS	SE LCS
0x70 – 0x7F	Allowed	Not allowed
Any other offset	Not allowed	Not allowed

Customer OTP Area description

There are 1024 bits assigned for Customer usable OTP.

Offset	Purpose	Notes
0x5F	Customer Security Flags	
0x60	Customer AES Key for Octal SPI (AES Key Slot0)	
0x64	Customer AES Key for Octal SPI (AES Key Slot1)	
0x68	General Purpose AES 128 for OEM use	
0x6C	General Purpose AES 128 for OEM use	
0x70	General OTP	OR Alt1 OEM RoT Public Key When Alt key flag set
0x74	General OTP	OR Alt1 OEM RoT Public Key When Alt key flag set
0x78	OEM General OTP bits	

Get MRAM data

Returns four words of MRAM data based on a provided MRAM word offset. The display shows the word offset entered followed by four words of data.

```
Select an option (Enter to return): 7
Enter hex word offset: 0
[0x0] 0x00000000 0x00000000 0x00000000 0x00000000
```

Trying to access ALIF protected MRAM will result in an error

```
Select an option (Enter to return): 7
Enter hex word offset: 0x57fff9
RX<-- length= 4 command= COMMAND_NAK          checksum= 0x10 error= ISP_ERROR_ILLEGAL_ADDRESS
[ERROR] 0x57fff9 ISP_ERROR_ILLEGAL_ADDRESS
```

Four words before the ALIF STOC area will also return an error.

This operation will not return any data if the device is in LCS=SE state.

Get log data

Returns the last SE-UART data sent:

```

.....
|  Name  |  CPU  |  Store Addr  |  Obj Addr  |  Dest Addr  |  Boot Addr  |  Size  |  Version  |  Flags  |  Time (ms)|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| DEVICE | CM0+  | 0x805C1EE0 | 0x805C14E0 | ----- | ----- | 672   | 0.5.0 | u V    | 0.00 |
| SERAM0 | CM0+  | ----- | 0x000000E0 | ----- | ----- | 89024 | 1.0.0 | ----- | 0.00 |
| SERAM1 | CM0+  | ----- | 0x00020AE0 | ----- | ----- | 89024 | 1.0.0 | ----- | 0.00 |
| HE_DBG | M55-HE | 0x805C2B80 | 0x805C2180 | 0x60000000 | 0x60000000 | 2256  | 1.0.0 | uLVB   | 0.00 |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
Legend: (u)(C)ompressed,(L)oaded,(V)erified,(s)kipped verification,(B)ooted,(E)ncrypted,(D)eferred

[SES] os Kernel V10.4.2
[SES] Main Task - looping forever...

```

Get trace data

Trace data are Markers left in SES memory space as SEROM and SES are executed. There are separate buffers used for SEROM and SES.

NOTE: If you are in RECOVERY Mode (SEROM only mode) you can only see the SEROM Trace data.

SEROM Trace Data

```
Select an option (Enter to return): 9
*** SEROM Trace Buffer decode ***
*** Trace Header ***
[0x00000100] Raw trace_vars_p: 0x0022232
No rollover detected
[0x00000104] 0xe0000000 (End Marker)
*****
```

Address	Raw Item	Seq#	LR	Trace Marker	Marker Data	CPU Freq	Diff(μs)	Time elapsed (μs)
[00000000]	0x026F0105	1	0x026F	Begin Reset_Handler	Data:0x01	76.8 MHz		
[00000004]	0x02783E0A	2	0x0278	Turn on SE System POWER Register	0x0000003F	76.8 MHz		
[00000008]	0x30290200	3	0x3029	SE Firewall configuration	Data:0x02	76.8 MHz		
[00000010]	0x30370311	4	0x3037	HOST Firewall configuration	Data:0x03	76.8 MHz		
[00000014]	0x4F350415	5	0x4F35	Firewall controller ready	Data:0x04	76.8 MHz		
[00000018]	0x028D0519	6	0x028D	Firewalls initialized	Data:0x05	76.8 MHz		
[0000001C]	0x0F71311E	7	0x0F71	SOC reset was triggered	0x00000001	76.8 MHz		
[00000024]	0x0F7D3122	8	0x0F7D	SOC reset was triggered	0x00000000	76.8 MHz		
[00000028]	0x0FA58526	9	0x0FA5	Configure EWIC Enable	0x0047FFC0	76.8 MHz		
[00000034]	0x0FB3862A	10	0x0FB3	Read EWIC Status	0x00000000	76.8 MHz		
[00000038]	0x50AD0F2D	11	0x50AD	Begin Main	Data:0x0F	76.8 MHz		
[00000040]	0x54DB8732	12	0x54DB	Read SOC_ID	0x0000B300	76.8 MHz		
[00000048]	0x54ED8836	13	0x54ED	override SOC_ID	0x0000B400	76.8 MHz		
[00000050]	0x2EBD0639	14	0x2EBD	Begin CC312 initializations	Data:0x06	76.8 MHz		
[00000054]	0x2EF1073D	15	0x2EF1	LCS = 0 -> CM	Data:0x07	76.8 MHz		
[00000058]	0x0AB71041	16	0x0AB7	Find STOC in MRAM	Data:0x10	76.8 MHz		
[0000005C]	0x0C7D2745	17	0x0C7D	BANK A is newer	Data:0x27	76.8 MHz		
[00000060]	0x50679049	18	0x5067	BANK A is VALID	Data:0x90	76.8 MHz		
[00000064]	0x5067934D	19	0x5067	BANK B is FIRST BOOT	Data:0x93	76.8 MHz		
[00000068]	0x11519651	20	0x1151	BOOT BANK A	Data:0x96	76.8 MHz		
[0000006C]	0x12B71155	21	0x12B7	Locate certificate chain	Data:0x11	76.8 MHz		
[00000070]	0x2DEF895A	22	0x2DEF	Certificate Active Software Version	0x10000000	76.8 MHz		
[00000078]	0x2E59895E	23	0x2E59	Certificate Active Software Version	0x20000000	76.8 MHz		
[00000080]	0x2E798962	24	0x2E79	Certificate Active Software Version	0xc0000000	76.8 MHz		
[00000088]	0x132F1465	25	0x132F	Verify certificate chain	Data:0x14	76.8 MHz		
[0000008C]	0x11FD196A	26	0x11FD	Begin certificate chain verification	0x00000003	76.8 MHz		
[00000094]	0x12111A6E	27	0x1211	Verify each certificate	0x905800C0	76.8 MHz		
[00000098]	0x12111A72	28	0x1211	Verify each certificate	0x90580408	76.8 MHz		
[000000A4]	0x12111A76	29	0x1211	Verify each certificate	0x90580750	76.8 MHz		
[000000AC]	0x127D1879	30	0x127D	End certificate chain verification	Data:0x18	76.8 MHz		
[000000B0]	0x118F987D	31	0x118F	BANK A BOOTED	Data:0x98	76.8 MHz		
[000000B4]	0x4F193282	32	0x4F19	Load MSP Address	0x3003FE00	76.8 MHz		
[000000BC]	0x4EFB3386	33	0x4EFB	Load JUMP Address	0x30009D85	76.8 MHz		
[000000C4]	0x0F491C89	34	0x0F49	Jump to SERAM	Data:0x1C	76.8 MHz		

SERAM Trace Data

```
Select an option (Enter to return): 10
*** SERAM Trace Buffer decode ***
*** Trace Header ***
[0x0000018c] Raw trace_vars_p: 0x001e1e3c
No rollover detected
[0x00000190] 0xe0000000 (End Marker)
*****
```

Address	Raw Item	Seq#	LR	Trace Marker	Marker Data	CPU Freq	Diff(μs)	Time elapsed (μs)
[00000000]	0x90B18E07	1	0x90B1	SERAM reset handler	T:580	76.8 MHz	7.55	7.55
[00000008]	0x3441050B	2	0x3441	Firewalls initialized	T:19939	76.8 MHz	252.07	259.62
[00000010]	0x99AF4F0F	3	0x99AF	main() start	T:32267	76.8 MHz	160.52	420.14
[00000018]	0x5EED8512	4	0x5EED	Configure EWIC Enable	0x007FFFC1	76.8 MHz		
[00000020]	0x5EFD8616	5	0x5EFD	Read EWIC Status	0x00000000	76.8 MHz		
[00000028]	0x95535A1B	6	0x9553	Cold boot	T:79703	76.8 MHz	617.66	1037.80
[00000030]	0xEEA76E1F	7	0xEEA7	Begin installing analog trim values	T:123634	76.8 MHz	572.02	1609.82
[00000038]	0xEEB56F23	8	0xEEB5	Finish installing analog trim values	T:124883	76.8 MHz	16.26	1626.08
[00000040]	0xEEBF7F26	9	0xEEBF	No trim values specified	0x00000000	76.8 MHz		
[00000048]	0x99534728	10	0x9953	Cc Lib END	T:269633	76.8 MHz		
[00000050]	0x95B1662F	11	0x95B1	Finish DCU configuration	T:271386	76.8 MHz	1884.77	3510.85
[00000058]	0x41554433	12	0x4155	Maintenance Detection BEGIN	T:272822	76.8 MHz	22.83	3533.67
[00000060]	0x41984537	13	0x4198	Maintenance Detection END	T:274214	76.8 MHz	18.70	3552.37
[00000068]	0xF0A1873A	14	0xF0A1	Read SOC_ID	T:274214	76.8 MHz	18.12	3570.49
[00000070]	0xF0B3883E	15	0xF0B3	Override SOC_ID	0x0000B300	76.8 MHz		
[00000078]	0x95034E42	16	0x9503	STOC Part#	0x0000B400	76.8 MHz		
[00000080]	0x0C6D8146	17	0x0C6D	SERAM boot error	0x32374541 AE72	76.8 MHz		
[00000088]	0x0C73814B	18	0x0C73	SERAM boot error	0x00000020	76.8 MHz		
[00000090]	0x96774C4F	19	0x9677	STOC DEVICE BEGIN	T:466073	76.8 MHz	2498.16	6068.66
[00000098]	0x96884D53	20	0x9688	STOC DEVICE END	T:1309459	76.8 MHz	10981.59	17050.25
[000000a0]	0x9D154257	21	0x9D15	PLL BEGIN	T:2812983	76.8 MHz	19577.14	36627.38
[000000a8]	0x9D43435B	22	0x9D43	PLL END	T:2818645	76.8 MHz	73.72	36701.11
[000000b0]	0x96E9555F	23	0x96E9	BANK Maintenance BEGIN	T:2940777	76.8 MHz	1590.26	38291.37
[000000b8]	0x96F35663	24	0x96F3	BANK Maintenance END	T:3529616	100 MHz	5888.39	44179.76
[000000c0]	0x973D5067	25	0x973D	STOC Process BEGIN	T:4258563	100 MHz	7289.47	51469.23
[000000c8]	0x9749526B	26	0x9749	TOC Process END	T:4259765	100 MHz	12.02	51481.25
[000000d0]	0x2123776F	27	0x2123	M55_HE released	T:4266483	100 MHz	67.18	51548.43
[000000d8]	0x97D85373	28	0x97D8	TOC Print BEGIN	T:4563003	100 MHz	2965.20	54513.63
[000000e0]	0x97FF5477	29	0x97FF	TOC Print END	T:5265934	100 MHz	7029.31	61542.94
[000000e8]	0x9828807B	30	0x9828	SERAM boot complete	T:22260013	100 MHz	169949.79	231492.73
					T:22833988	100 MHz	5730.75	237223.48

Get Power data

This option returns the Power state of the SoC:

```
Select an option (Enter to return): 11
es0_ppu_status      0x00000000    OFF
es1_ppu_status      0x00000008    ON
se_ppu_status        0x00000008    ON
fw_ppu_status        0x00000007    FUNC_RET
systop_ppu_status    0x00000008    ON
dbgtop_ppu_status    0x00000008    ON
clustop_ppu_status   0x00000000    OFF
a32_0_ppu_status     0x00000000    OFF
a32_1_ppu_status     0x00000000    OFF
modem_ppu_status     0x00000000    OFF
modem_aon_status     0x00000000    OFF
sse700_aon_status    0x00000001    ON
```

NOTE: If you are in Hard Maintenance mode these values will not be accurate as the state changes in this mode.

Get Clock data

This option returns the Clock and PLL state of the SoC:

```
12 - Get clock data

Select an option (Enter to return): 12
Clock Status
  HFXTAL STARTED
  PLL LOCKED
  SE CLOCK: PLL
CLK freq A32 800Mhz
CLK freq ES0 400Mhz
CLK freq ES1 160Mhz
SE frequency 100.03MHz

Registers:
HOSTCPUCLK_CTRL : 0x00000004
HOSTCPUCLK_DIV0 : 0x001F0000
HOSTCPUCLK_DIV1 : 0x001F0000
ACLK_CTRL       : 0x00000202
ACLK_DIV0       : 0x00000000
OSC_CTRL        : 0x00110011
PLL_LOCK_CTRL   : 0x00000001
PLL_CLK_SEL     : 0x00110111
ESCLK_SEL       : 0x00000033
CLK_ENA        : 0x11033111
SYSTOP_CLK_DIV  : 0x00000102
MISC_REG1       : 0x000002A4
XO_REG1         : 0x11D08439
  xtal_cap:8  gm_pfet:16  gm_nfet:16
PD4_CLK_SEL     : 0x00000000
PD4_CLK_PLL     : 0x00000000
MISC_CTRL       : 0x00001001
DCDC_REG1       : 0x614DE693
DCDC_REG2       : 0x8F014441
VBAT_ANA_REG1   : 0x06479E80
  osc_rc_32k_freq_cont:0  xtal32k_en:1  xtal32k_gm_cont:15  xtal32k_cap_cont:8  bor_en:1  bor_hyst:1  bor_thresh:0
VBAT_ANA_REG2   : 0x00C06B6
  pmubg_vref_cont:11  osc_76Mrc_cont_bit0:1  osc_76M_div_cont_fast:0  osc_76Mrc_cont:16  osc_76M_div_cont_slow:1
```

MRAM

Top level menu group

```
Select an option (Enter to exit): 3

Available options:

1 - Erase Application Mram
2 - Fast Erase Application Mram
3 - Fast Erase App. Mram (include NTOC)
4 - Get MRAM info

Select an option (Enter to return):
```

NOTE: If you are in Recovery (ROM) mode, these commands will not operate as SEROM does not support these MRAM erase commands.

Erase Application MRAM

```
Select an option (Enter to return): 1
```

Name	CPU	Store Addr	Obj Addr	Dest Addr	Boot Addr	Size	Version	Flags	Time (ms)
DEVICE	CM0+	0x8057cb90	0x8057C190	-----	-----	296	0.5.0 u V		15.44
DEVICE	CM0+	0x805c1ec0	0x805C14C0	-----	-----	340	1.0.0 u V		15.74
* SERAM0	CM0+	-----	0x000000C0	-----	-----	83408	1.101.0 u s		0.00
SERAM1	CM0+	-----	0x00020AC0	-----	-----	83408	1.101.0 -----		0.00
BLINK-HE	M55-HE	0x8057d6c0	0x8057CCC0	0x58000000	0x58000000	10440	1.0.0 uLVB		16.92

Available options:

This option will erase the Application MRAM. It is the equivalent of using the ‘-e app’ option of app-write-mram.

In this example, we see that we have one Application (BLINK-HE) running from Application MRAM.

We will now erase this. Choose the Erase Application MRAM. This will instruct SES to erase the Application MRAM. The size of the MRAM to erase is known as you selected the device type using the tools-config.

```
Available options:

1 - Erase Application Mram
2 - Fast Erase Application Mram
3 - Get MRAM info

Select an option (Enter to return): 1
[INFO] erasing 0x80000000 5,767,168 bytes
[INFO] Full Erase done

Available options:

1 - Erase Application Mram
2 - Fast Erase Application Mram
3 - Get MRAM info

Select an option (Enter to return):
```

If we inspect the TOC information, we can see that the BLINK-HE application has been removed from MRAM.

Select an option (Enter to return): 1

Name	CPU	Store Addr	Obj Addr	Dest Addr	Boot Addr	Size	Version	Flags	Time (ms)
DEVICE	CM0+	0x805c1ec0	0x805c14c0	-----	-----	340	1.0.0 u V		15.74
* SERAM0	CM0+	-----	0x000000c0	-----	-----	83408	1.101.0 u s		0.00
SERAM1	CM0+	-----	0x00020Ac0	-----	-----	83408	1.101.0 -----		0.00

Fast Erase Application MRAM

This option will invalidate the Application TOC without erasing the Application MRAM. While the effect will be the same as the Erase Application MRAM option, this option does not *clear* the Application MRAM area.

Fast Erase Application MRAM (including NTOC)

This option will invalidate the Application TOC and erase the first 16 bytes of Application MRAM.

NTOC is the method by which the M55_HE boots from the start of MRAM (address 0x80000000)

This option is needed when the user wants to invalidate a binary previously burnt in MRAM address 0x80000000, otherwise the ATOC has been erased but SES will still find a valid NTOC and boot it.

Get MRAM info (MRAM Walker)

This walks the entire MRAM memory looking for Table of Contents objects:

```
Available options:
1 - Erase Application Mram
2 - Fast Erase Application Mram
3 - Get MRAM info

Select an option (Enter to return): 3
ATOC 0x8057ffc0
+ header      OEMTOC01
+ header_size 16
+ # toc entries 1
+ entry_size  32
+ version     0x1
STOC 0x80580000
+ header      ALIFTOC1
+ header_size 48
+ # toc entries 4
+ entry_size  32
+ version     0x1
ATOC 0x8058eca0
ATOC 0x805af6a0
```

In this example, two System TOC objects were found in MRAM.

Utilities

Top level menu group

```
Available options:

1 - Terminal mode
2 - Get SERAM metrics
3 - Get ECC key
4 - Get Firewall configuration
5 - Alt Boot OTP

Select an option (Enter to return): |
```

NOTE: Devices that can support Alternative Boot path from external memory will show Option#5.

Terminal Mode

This option echoes the SE-UART output.

```
Select an option (Enter to return): 1
[TERMINAL] Ctrl-C to exit

SEROM v1.96.0 0x0000B400

SES B4 v1.103.0 Dec 12 2024 20:07:29
[SES] No ATOC
[SES] STOC DEVICE ok
[SES] No LF XTAL

[SES] SERAM bank 0x0 is valid and booted
[SES] STOC ok
[SES] M55-HE booted from address 0x58000000
[SES] LCS=1
[SES] FC:Rgn
0:2 7:0 8:0 9:0 13:0 13:1 13:2
```

Name	CPU	Store Addr	Obj Addr	Dest Addr	Boot Addr	Size	Version	Flags	Time (ms)
DEVICE	CM0+	0x805C1EC0	0x805C14C0	-----	-----	340	1.0.0	u V	15.83
* SERAM0	CM0+	-----	0x000000C0	-----	-----	86196	1.103.0	-----	0.00
SERAM1	CM0+	-----	0x00020AC0	-----	-----	86196	1.103.0	-----	0.00

```
Legend: (u)(C)ompressed, (L)oaded, (V)erified, (s)kipped verification, (B)ooted, (E)ncrypted, (D)eferred

[SES] SE frequency is 102.99 MHz
```

In this example, [RESET] was pressed on the board, and we see the complete boot up information echoed.

To exit this mode press CTRL-C and it will drop back to the maintenance menu.

SES Metrics

```
Available options:

1 - Terminal mode
2 - Get SERAM metrics

Select an option (Enter to return): 2
```

TaskName	Size	Used
SERAM_main_task	1024	300
SERAM_uart_task	3072	720

```
SES uptime 0:00:04:20
```

Returns (for now) stack information details used by SES as well as the uptime since SES booted. The uptime format is Days:Hours:Minutes:Seconds

Get address/Set address

These diagnostic commands are not supported.

Get ECC key

This option returns the public ECC key of the device. If the ECC key is not provisioned yet, all 0's is returned.

```
Available options:
1 - Terminal mode
2 - Get SERAM metrics
3 - Get ECC key
4 - Get Firewall configuration

Select an option (Enter to return): 3
ECC key (HEX): 26E6434817CE4227585BE3068A05322196005A5349E2D1A7A9D9AE8185DDE76DC5C20DB614A64227AB218D514E76F248EEFD4B46C4AF87602C123A08FF56E5AF
```

Get Firewall configuration

This option returns the current Firewall configuration status, i.e., what regions have been configured in which Firewall Components (FC: x').

```
FC: 3 region: 2
FC: 3 region: 3
FC: 3 region: 4
FC: 3 region: 5
FC: 3 region: 6
FC: 3 region: 7
FC: 3 region: 8
FC: 3 region: 9
FC: 3 region: 10
FC: 3 region: 11
FC: 3 region: 12
FC: 3 region: 13
FC: 3 region: 14
FC: 3 region: 15
FC: 3 region: 16
FC: 3 region: 17
FC: 3 region: 18
FC: 3 region: 19
FC: 3 region: 20
FC: 3 region: 24
FC: 3 region: 25
FC: 3 region: 26
FC: 3 region: 27
FC: 3 region: 28
FC: 3 region: 29
FC: 3 region: 32
FC: 3 region: 33
FC: 3 region: 39
FC: 4 region: 0
FC: 4 region: 1
FC: 5 region: 0
FC: 5 region: 1
```

Alternate Boot OTP

This option programs the OTP bits required to support Alternative boot path from External memory.

When executed, it generates OTP information based on configuration from *build/config/otp-altboot.json* file and sends it to write the corresponding OTP area.

```
Available options:
1 - Terminal mode
2 - Get SERAM metrics
3 - Get ECC key
4 - Get Firewall configuration
5 - Alt Boot OTP

Select an option (Enter to return): 5
*****
!!!WARNING!!! This action will permanently update OTP memory
*****
Please enter the following confirmation string followed by [ENTER] to continue,
or just press [ANY] key to abandon this action
*****
Confirmation string: NEHZXQ
> NEHZXQ
[INFO] OTP will be updated. Enter Yes to continue (or any key to abort)
> yes
```

There are some confirmations to complete this command as you are writing to OTP, and this is a one-way operation; you **cannot undo these changes**.

Once the command is executed, you can retrieve revision information from the device.

```

select an option (Enter to return): 5
Source      =  SERAM
Version     =  0x02A0
ALIF_PN     =  0x0
HBK0       =  00000000000000000000000000000000
HBK1       =  00000000000000000000000000000000
HBK_Fw     =  00000000000000000000000000000000
Wounding    =  0x0
DCU         =  0x0
MfgData     =  0000000000000000000000000000000000000000000000000000000000000000
LCS         =  0x0 (CM)
SES Alt Load = 00000100
      Loading options      : 0x1 MRAM then Ext Memory
      Ext Memory type     : 0x0 OSPI Flash 8bit in XIP Mode

SES Ext Conf = 000B0B0B0105E22B8ADF2B0E000000
      rx_sample_delay      : 0x0
      xip_incr_inst        : 0x8
      xip_wrap_inst        : 0x8
      aes_rxd_s_delay      : 0x8
      ddr_drive_edge       : 0x1
      pinmux_sel           : 0x5
      bus_speed_sel        : 0x0
      fast_read_wait_cycles: 0x8
      flash_size           : 0x0      32 MB
      valid                : 0x1
      reset_port           : 0xF
      reset_pin            : 0x5
      spi_mode             : 0x0
      io_mode              : 0x2
      ddr_en               : 0x0
      xip_ddr_en           : 0x1
      inst_ddr_en          : 0x0
      xip_inst_ddr_en      : 0x0
      rxds_en              : 0x0
      rxds_vl_en           : 0x1
      dfs                  : 0xF
      xip_dfs              : 0xF
      xip_rxds_en          : 0x1
      xip_instr_len        : 0x2
      xip_addr_len         : 0x8
      xip_spi_mode         : 0x3
      xip_hyperbus_en      : 0x0
      tx_wait_count        : 0x0
  
```

Here there are more fields printed showing the values set for the external memory.

Setting capabilities

Top level menu group

```

Available options:

1 - Enable LOGGING
2 - Disable LOGGING
3 - Enable PRINTING
4 - Disable PRINTING
  
```

Enable / Disable LOGGING

Controls whether the SES messages are added to the log buffer. The log buffer can be read using the 'Device Information' option.

Enable / Disable PRINTING

Toggle PRINTING – controls whether the SERAM messages are printed directly to the SE UART or put in a buffer from where they are periodically printed out.

ROM

Devices support a limited subset of ISP commands built into the ROM.

If the NVM contains no viable SES image to load, SEROM will enter an ISP mode to allow recovery.

This Recovery mode will take an SES image and write it to the NVM using a specific ISP protocol supported by SEROM, it is not the same as used for writing the Application MRAM.

Top level menu group

```
Available options:
1 - ROM
2 - Device Information
3 - Utilities
4 - Alt. Boot Octal SPI
Select an option (Enter to exit): |
```

Alt. Boot Octal SPI

The following SEROM based OSPI operations are available:

```
Available options:
1 - ROM
2 - Device Information
3 - Utilities
4 - Alt. Boot Octal SPI
Select an option (Enter to exit): 4
Available options:
1 - OSPI erase sector size 4K
2 - OSPI erase sector size 32K
3 - OSPI erase sector size 128K
4 - OSPI erase chip
5 - OSPI flag toggle
6 - OSPI settings
7 - OSPI init
```

These options allow a user to configure OSPI for External Memory boot.

OSPI erase sector size

Erases the OSPI memory for various sizes.

OSPI erase chip

Erases the OSPI memory

OSPI flag toggle

Internal use only

OSPI settings

Uses the binary file generated from Maintenance tool Alternate Boot OTP option. This does NOT burn the OTP but simply sends the OSPI settings to the device

OSPI init

Initializes the OSPI using the settings sent to the device or the default ones stored in SEROM.

Recovery Session – How do I know I am in Recovery Mode?

A device in recovery mode will respond to Maintenance as follows:

```
[INFO] COM11 open Serial port success
[INFO] baud rate 57600
[INFO] Connecting to target...Device connected in Recovery

Available options:

1 - ROM
2 - Device Information
3 - Utilities
4 - Alt. Boot Octal SPI

Select an option (Enter to exit): |
```

Maintenance attempts to connect to the Target device and in this case discovered the device is in Recovery Mode.

The number of Menu options presented now in Maintenance is less than before as there are limited operations allowable when in Recovery Mode.

Recovery Session – MRAM

Select the ROM -> Recovery option to start the recovery to MRAM.

```
Available options:
1 - Recovery
2 - Recovery (No Reset)
3 - Recovery OSPI (No Reset)

Select an option (Enter to return): 1
Bootloader stage: SEROM
Detected Part#: AE822FA0E5597LS0
Detected Revision: A0
[INFO] System TOC Recovery with parameters:
- Device Part# E8 (AE822FA0E5597LS0) - 5.5 MRAM / 9.75 SRAM - Rev: A0
- ALIF Base Address: 0x80580000
- Package: alif/SP-AE822FA0E5597LS0-rev-a0-dev.bin
- Offset: alif/offset-58-rev-a0-dev.bin
[INFO] writing main MRAM image 'alif/SP-AE822FA0E5597LS0-rev-a0-dev.bin' to address 0x580000
alif/SP-AE822FA0E5597LS0-rev-a0-dev.bin[#####]100%: 270400/270400 bytes
[INFO] recovery time: 269.95 seconds
[INFO] writing offset 'alif/offset-58-rev-a0-dev.bin' to address 0x5ffff0
alif/offset-58-rev-a0-dev.bin [#####]100%: 16/16 bytes
[INFO] recovery time: 0.02 seconds
[INFO] Target reset
Recovery process finished. Please reload maintenance tool for verification
```

The device is in recovery mode and has loaded the SES image via SEROM into MRAM.

Recovery Session - OSPI

```
Available options:
1 - Recovery
2 - Recovery (No Reset)

Select an option (Enter to return): 1
Bootloader stage: SEROM
Bring Up mode - Blank part detected!
Detected Part#: AE722F80F55D5LS
Detected Revision: B4
Device is not provisioned!
[INFO] System TOC Recovery with parameters:
- Device Part# E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B4
- MRAM Base Address: 0x80580000
alif\SP-AE722F80F55D5LS-rev-b4-dev.bin[#####]100%: 270368/270368 bytes
[INFO] recovery time 110.68 seconds
alif\offset-58-rev-b4-dev.bin [#####]100%: 16/16 bytes
[INFO] recovery time 0.00 seconds
[INFO] Target reset
```

This feature is only available on ENSEMBLE E4, E6, E8 Revision REV_A1

The device is in recovery mode and has loaded the SES image via SEROM into External OSPI memory.

OSPI options not set

For External memory booting to work the device must be provisioned with OSPI details in OTP. If this is not done, then recovery to OSPI is not possible.


```
Available options:
1 - Recovery
2 - Recovery (No Reset)
3 - Recovery OSPI (No Reset)

Select an option (Enter to return): 3
[WARN] No Part# was detected! Defaulting to AE822FA0E5597LS0
[WARN] Device is not provisioned!
Bootloader stage: SEROM
Detected Part#: AE822FA0E5597LS0
Detected Revision: FPGA_A1
Connected target is not the default Revision
Do you want to set this part as default? (y/n): y
[INFO] OSPI Recovery requested
[INFO] Device supports OSPI external memory
[INFO] OSPI is NOT enabled in OTP

[ERROR] OSPI is not enabled in OTP
```

No Reset

By default, after the recovery operation is performed then the tool will send a request to perform a RESET of the platform.

There is also an option to NOT perform this reset after recovery. This can be useful for observing potential failures that would be masked by the actual reset operation.

An example of this would be a device failing with say a CC312 error code (0x0B000006 - illegal KCEICV) because of the automatic reset, and the user resetting the board one more time after that, both SERAM banks are being marked as invalid, so the reported error code is 0x26 in the recovery screen, which hides the actual issue.

Target not in recovery mode

```
Available options:
1 - Device Control
2 - Device Information
3 - MRAM
4 - Utilities
5 - Setting capabilities
6 - ROM

Select an option (Enter to exit): 6

Available options:
1 - Recovery
2 - Recovery (No Reset)

Select an option (Enter to return): 1
Bootloader stage: SERAM
[ERROR] Device not in Recovery mode, use updateSystemPackage Tool
```

This message is shown if the target device is not in recovery mode probably because it is running SES already.

Target device mismatch

```
Available options:
1 - Device Control
2 - Device Information
3 - MRAM
4 - Utilities
5 - Setting capabilities
6 - ROM

Select an option (Enter to exit): 6

Available options:
1 - Recovery
2 - Recovery (No Reset)

Select an option (Enter to return): 1
Bootloader stage: SEROM
Detected Part#: AE722F80F55D5LS
Detected Revision: B4
Connected target is not the default Part#
Connected target is not the default Revision
Do you want to set this part as default? (y/n): |
```

This message shows that the target Device is different from the one selected by tools-config for this session.

Recovery UART Speed

When in RECOVERY mode the device is running ISP from the ROM which runs at RC frequency, so you will notice it is slower than when SES is performing MRAM updates.

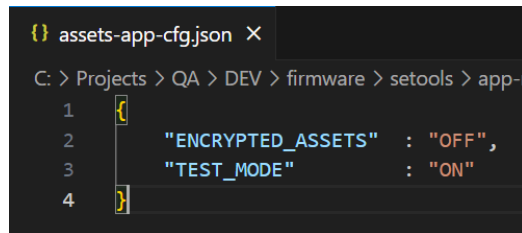
- There is no PLL support inside the ROM code so the baud rate cannot be increased like it is when SES is doing updates (Using app-mram-write, updateSystemPackage).
- ISP protocol used in the ROM is MRAM line based i.e. 128bits at a time with no buffering.

app-assets-gen

The app-assets-gen tool generates the “assets package” required to provision the device with the app-provision tool (see next section).

This file is loaded by the app-provision tool and contains assets and options for the provisioning. Provisioning means keys and other values to be programmed (OTP) into the device.

The options can be configured with a JSON file. Default configuration file is `build\config\assets-app-cfg.json`



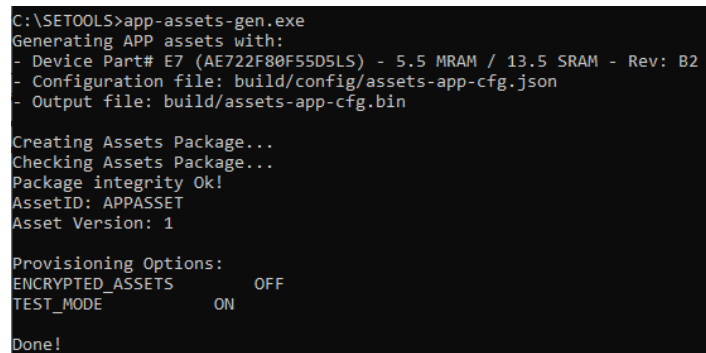
```
{
  "ENCRYPTED_ASSETS" : "OFF",
  "TEST_MODE"       : "ON"
}
```

Options description:

- **ENCRYPTED_ASSETS:**
 - Defines if the assets are encrypted or not. **Unencrypted assets should only be used in trusted environments.** Check out the next section to learn how to generate encrypted assets.
- **TEST_MODE:**
 - This option DOES NOT burn the OTP area, so the program can run multiple times without advancing the LCS to SE.
 - This is provided to test the process without affecting the device.

Once the desired configuration is done, the assets package can be generated using the tool:

```
$ app-assets-gen (-h for help)
```



```
C:\SETTOOLS>app-assets-gen.exe
Generating APP assets with:
- Device Part# E7 (AE722F80F5D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B2
- Configuration file: build/config/assets-app-cfg.json
- Output file: build/assets-app-cfg.bin

Creating Assets Package...
Checking Assets Package...
Package integrity Ok!
AssetID: APPASSET
Asset Version: 1

Provisioning Options:
ENCRYPTED_ASSETS      OFF
TEST_MODE            ON

Done!
```

The command generates the assets package and checks the generated binary to extract the Options selected (it reads back the options from the binary, not from the json file).

Also, using the `-c` flag, the tool will skip the assets package generation, and it just checks the options used to generate it.

```
$ app-assets-gen -c
```

```
C:\SETTOOLS>app-assets-gen.exe -c
Generating APP assets with:
- Device Part# E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B2
- Output file: build/assets-app-cfg.bin

Checking Assets Package...
Package integrity Ok!
AssetID: APPASSET
Asset Version: 1

Provisioning Options:
ENCRYPTED_ASSETS      OFF
TEST_MODE            ON

Done!
```

Using the `-f` option, we can specify an input file (.json format if assets package is being generated, or .bin file if only checking the options used to generate the binary).

Example:

```
$ app-assets-gen -f build\config\assets-cfg.json
$ app-assets-gen -c -f build\assets-cfg.bin
```

The output file is always created in the `build\` folder and the name will be the same as the configuration file.

Procedure to encrypt assets

Encrypted assets should always be used in **untrusted environments** (i.e., when a third party performs the provisioning). This ensures that assets remain protected, as they are stored in encrypted form within the file sent to another party.

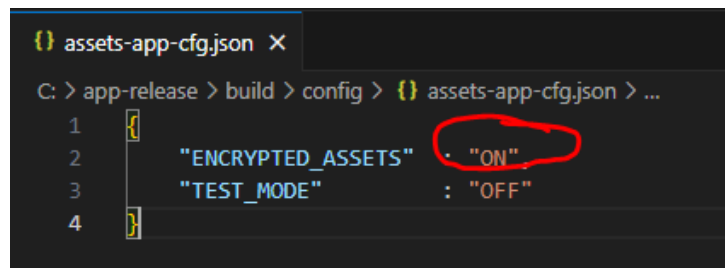
The assets are encrypted using a **derived key (Koem)**, which is generated from a 128-bit AES key embedded in the chip design (RTL), known as the **Platform Key (Krtl)**. This derivation follows **NIST SP 800-108** guidance and uses the **Root of Trust (RoT)** of the OEM.

Since the **Krtl key** is an asset owned by **Alif Semiconductor**, the OEM must submit a request to Alif. Alif will then derive the **Koem key** and send it back to the OEM.

To protect the **Koem** key during exchange between Alif and the OEM, the **OEM Request Package** (formatted as a certificate) contains the **public key** of a temporary RSA key pair generated specifically for this purpose. Upon receiving the request, Alif extracts the temporary public key, encrypts the **Koem**, and returns it to the OEM as the **ICV Response Package**.

When the OEM receives this response package, it uses the **temporary private key** to decrypt it and retrieve the original **Koem** derived by Alif. This key is then used by **SETOOLS** to encrypt OEM's assets.

To configure encrypted assets, set the "ENCRYPTED_ASSETS" attribute "ON", in the *build/config/assets-app-cfg.json* file.



```
{
  "ENCRYPTED_ASSETS": "ON",
  "TEST_MODE": "OFF"
}
```

To generate the OEM Request Package, run the app-assets-gen tool. This process will generate the OEM Request Package file (utils/key/oem-request_pkg.bin);

```
C:\Windows\System32\cmd.exe

C:\app-release>python3 app-assets-gen.py
Generating APP assets with:
- Device Part# E8 (AE822FA0E5597LS0) - 5.5 MRAM / 9.75 SRAM - Rev: A0
- Configuration file: build/config/assets-app-cfg.json
- Output file: build/assets-app-cfg.bin

Generating Temporary key pair (to generate ICV Request package)
2025-10-13 19:37:22,426 - RSA Key Generator Utility started (Logging to build/logs/key_gen_log.log)
2025-10-13 19:37:22,562 - Private key written to: utils/key/oem_tmp_enc_key.pem
2025-10-13 19:37:22,563 - Public key written to: utils/key/oem_tmp_enc_pub_key.pem
2025-10-13 19:37:22,564 - Script completed successfully
Merging key pairs...
Generating utils/key/OEMRoT_key_pair.pem file...
Generating utils/key/oem_tmp_enc_key_pair.pem file...
Generating Request package...
2025-10-13 19:37:22,578 - OEM key request Utility started (Logging to build\logs\key_request_cert.log)
2025-10-13 19:37:22,579 - Parsing config file: utils\cfg\dmpu_oem_key_request.cfg
2025-10-13 19:37:22,582 - **** Generate OEM key requesting certificate ****
2025-10-13 19:37:22,836 - **** OEM key request generation has been completed successfully ****

*****
The request package has been generated: utils/key/oem_request_pkg.bin
Please send it to Alif to get the response package
*****

C:\app-release>
```

Send this **OEM Request Package** to Alif (or via the Support Portal) to get the **ICV Response Package**. Once received, store the file in the same directory as the request file (*utils/key/* folder). Continue the process running again the *app-assets-gen.py* tool.

```
C:\Windows\System32\cmd.exe

C:\app-release>python3 app-assets-gen.py
Generating APP assets with:
- Device Part# E8 (AE822FA0E5597LS0) - 5.5 MRAM / 9.75 SRAM - Rev: A0
- Configuration file: build/config/assets-app-cfg.json
- Output file: build/assets-app-cfg.bin

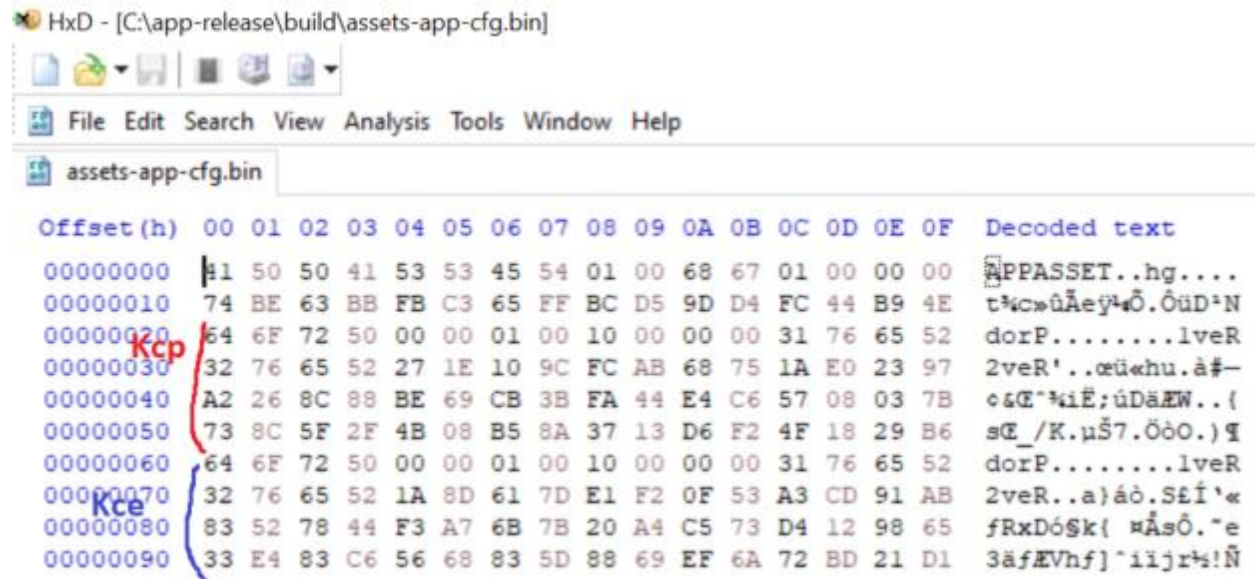
Processing ICV Response package...
2025-10-13 19:53:00,567 - OEM asset packaging Utility started (Logging to build/logs/oem_asset_pkg.log)
2025-10-13 19:53:00,568 - Parsing config file: utils/cfg/asset_oem_ce.cfg
2025-10-13 19:53:00,625 - **** Generate DMPU Asset package ****
2025-10-13 19:53:00,775 - **** OEM Asset package generation has been completed successfully ****
2025-10-13 19:53:00,776 - OEM asset packaging Utility started (Logging to build/logs/oem_asset_pkg.log)
2025-10-13 19:53:00,777 - Parsing config file: utils/cfg/asset_oem_cp.cfg
2025-10-13 19:53:00,781 - **** Generate DMPU Asset package ****
2025-10-13 19:53:00,923 - **** OEM Asset package generation has been completed successfully ****
Creating Assets Package...
Checking Assets Package...
Package integrity Ok!
AssetID: APPASSET
Asset Version: 1

Provisioning Options:
ENCRYPTED_ASSETS    ON
TEST_MODE          OFF

Done!

C:\app-release>
```

The assets package (assets-ap-cfg.bin) will be generated in the build/ folder. Open the file with a hex editor, and check that both Kce and Kcp assets are now encrypted.



app-provision

The app-provision tool will inject user assets (keys and secrets) into the device and move the device from DM (Device Manufacturing) to SE (Secure Enable) LCS (Life-Cycle State).

The SE LCS is used for devices from the manufacturing line and “in the field”. It permits the execution of security functions but blocks all debugging and testing capabilities.

Using Secure Boot is mandatory in this SE LCS.

```
$ app-provision (-h for help)
```

```
C:\SETTOOLS>app-provision.exe
APP Provision with parameters:
Device Part# E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B2
Assets file: build/assets-ap-cfg.bin

[INFO] COM12 open Serial port success
[INFO] Running APP Provisioning code...
←[94m APP Provision ran in TEST MODE!
←[0m
[INFO] Done
```

The default assets file is build\assets-ap-cfg.bin. Use the -a option to specify other assets file.

To perform a real provision, configure TEST_MODE OFF in the assets configuration file, and re-build the assets using the app-assets-gen tool;

```
C:\SETTOOLS>app-assets-gen.exe
Generating APP assets with:
- Device Part# E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B2
- Configuration file: build/config/assets-app-cfg.json
- Output file: build/assets-app-cfg.bin

Creating Assets Package...
Checking Assets Package...
Package integrity Ok!
AssetID: APPASSET
Asset Version: 1

Provisioning Options:
ENCRYPTED_ASSETS      OFF
TEST_MODE            OFF

Done!
```

Running the app-provision tool with this new assets' binary, it will provision the device;

```
C:\SETTOOLS>app-provision.exe
APP Provision with parameters:
Device Part# E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B2
Assets file: build/assets-app-cfg.bin

[INFO] COM12 open Serial port success
[INFO] Running APP Provisioning code...
←[94m APP Provision Return Code: 0x0
←[0m
[INFO] Done
```

Once this script finishes, the device will advance from DM to SE (Secure Enable) LCS, after reset.

```
C:\Windows\System32\cmd.exe - maintenance.exe
Select an option (Enter to return): 1
[TERMINAL] Ctrl-C to exit

SEROM v0.47.68 0x0000B200

[SES] Cold boot path
*** Host Firewall configured

[SES] MRAM error bypass is Enabled

SES B0 EVALUATION_BOARD v1.0.87 Dec 8 2023 23:59:45
[SES] Device ID = 0x0000B200
[SES] PLL code version 0.0.4
[SES] LCS=5
[SES] System partition address 0x80580000
[DEV] Wounding Data: 0x00C0FFFB
[SES] System device configuration processed (0x00000000) BL_STATUS_OK
[SES] Application device configuration processed (0x00000001) BL_ERROR_APP_INVALID_TOC_ADDRESS
[SES] System partition processed (0x00000000) BL_STATUS_OK
[SES] Application partition processed (0x00000001) BL_ERROR_APP_INVALID_TOC_ADDRESS
```

app-secure-debug (--rma)

A device in SE LCS (Secure State) blocks all debugging and testing capabilities. The Secure Debug tool gives the user a mechanism to:

- Open debugger access to the APP cores and the related debugging infrastructure, in a device in SE LCS.
- Initiate the process of transitioning a device to RMA LCS (end of product cycle), before sending the device to Alif for failure analysis or further inspection.

Both operations are done by generating a secure debug certificate and sending it to the device via ISP. The secure debug certificate is signed using the same keys as the ones used when generating the ATOC, i.e. for secure booting. The certificates are session-specific and become invalid after the device is reset. SES requires that a secure debug request is signed, regardless of whether the request is for secure debug unlocking or for the RMA. SES will verify the signed request before continuing.

To open debugger access to the APP cores, run:

```
$ app-secure-debug (-h for help)
```



```
C:\Windows\System32\cmd.exe

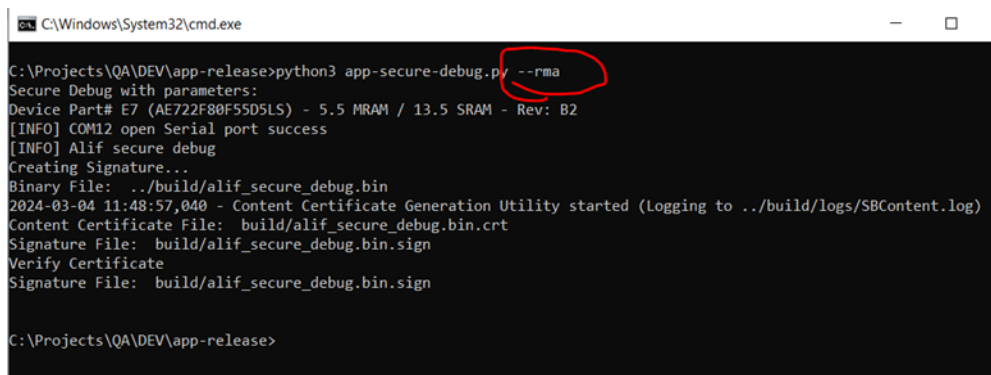
C:\Projects\QA\DEV\app-release>python3 app-secure-debug.py
Secure Debug with parameters:
Device Part# E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B2
[INFO] COM12 open Serial port success
[INFO] Alif secure debug
Creating Signature...
Binary File: ../build/alif_secure_debug.bin
2024-03-04 11:43:55,787 - Content Certificate Generation Utility started (Logging to ../build/logs/SBContent.log)
Content Certificate File: build/alif_secure_debug.bin.crt
Signature File: build/alif_secure_debug.bin.sign
Verify Certificate
Signature File: build/alif_secure_debug.bin.sign

C:\Projects\QA\DEV\app-release>
```

After this command, a debugger can connect to the application cores and do a debug session.

To initiate the transition to RMA LCS, run:

```
$ app-secure-debug --rma
```



```
C:\Windows\System32\cmd.exe

C:\Projects\QA\DEV\app-release>python3 app-secure-debug.py --rma
Secure Debug with parameters:
Device Part# E7 (AE722F80F55D5LS) - 5.5 MRAM / 13.5 SRAM - Rev: B2
[INFO] COM12 open Serial port success
[INFO] Alif secure debug
Creating Signature...
Binary File: ../build/alif_secure_debug.bin
2024-03-04 11:48:57,040 - Content Certificate Generation Utility started (Logging to ../build/logs/SBContent.log)
Content Certificate File: build/alif_secure_debug.bin.crt
Signature File: build/alif_secure_debug.bin.sign
Verify Certificate
Signature File: build/alif_secure_debug.bin.sign

C:\Projects\QA\DEV\app-release>
```

After this command, the device is still in its current LCS (SE or DM) but the RMA transition is initiated. The device can then be sent to Alif, where the transition to RMA LCS will be completed.

Tool Exit codes

The tools will return the following runtime execution Error codes.

- 0 Indicates success.
- 1 Indicates failure.

NACK Error Handling

The ISP protocol uses an Acknowledge (ACK) / Not Acknowledge (NACK) protocol scheme. If a NACK is seen, then it will be followed by an Error code.

If a NACK packet is seen it will always be printed, and the tool will EXIT.

```
$ python3 updateSystemPackage.py -m isp -x
Burning: System Package in MRAM
Device Part# AE722F80957D2CH - Rev: A0
- MRAM Base Address: 0xa0580000

alif\header.bin          [#####]100%: 48/48 bytes
alif\SystemPackage.bin   [#####]100%: 281280/281152 bytes
RX<-- length= 4 command= COMMAND_NAK      checksum= 0x6 error= ISP_BAD_DEST_ADDRESS
```

TIMEOUT Error Handling

An ISP session requires an initial ISP START command sent to the Target to put it into ISP mode. If this command fails, then the ISP session will not occur, and an Error is reported that the Target did not respond

```
Available options:

1 - maintenance mode
2 - device reset
3 - device enquiry
4 - get revision info
5 - get TOC info

Select an option: 4
[ERROR] Target did not respond
```

This Error will exit the program.

Possible causes for this Error:

- Incorrect baud rate setting
- Target is not powered or is in a low power state such as STOP mode.

10. RECOVERY MODE

Recovery mode occurs when SEROM

- Fails to find an SES image in MRAM that it can boot.
- Fails to verify an SES image from MRAM.
- Fails to decrypt an SES image from MRAM.
- Finds the dual SES Banks are both marked as “bad”.

If your device is in Recovery mode, you will see the following SE-UART output

```
[TERMINAL] Ctrl-C to exit
SEROM v1.93.1 0x0000A001
[SEROM] BOOT_LOADER_find_and_load_seram(): error: 0x00000030; extended error code: 0x00000000
[SEROM] LCS = 0x0

[SEROM][ 0x7A606000 ] MRAM_CONTROLLER_BASE_ADDRESS = 0x00000000
[SEROM][ 0x7A606004 ] MRAM_CONTROLLER_FSM_STATE = 0x00000000
[SEROM][ 0x7A606008 ] MRAM_CONTROLLER_ERROR_CAUSE = 0x00000000
[SEROM][ 0x7A60600C ] MRAM_CONTROLLER_PENDING_0 = 0x00000000
[SEROM][ 0x7A606010 ] MRAM_CONTROLLER_PENDING_1 = 0x00000000

Dump trace buffer
0x037B0105 0x03873E0A 0x0000003F 0x35E5020D 0x35F30311 0xC91D0415 0x03990519 0x10CD311E 0x00000001 0x10D93122
0x00000000 0x11018526 0x001FFFFF 0x110F862A 0x00000000 0xCD210F2D 0x34790631 0x34AD0735 0x0C5F1039 0x0BE3343D
0x62991D42 0x00000030 0x62A33646 0x00000000 0xD99DFE09 0x4AD85CEB 0x978CB4D4 0x337A8FA3 0x66E0781B 0xFFB5BEEF
0xBF8DC6D8 0x0BEAED88 0x9DEAA2BC 0xFCDFC46E 0x73E7276E 0x1C2FE577 0xFA7E653B 0xEF7F7132 0x3F7FF876 0x2B7207A9
0xCDE88D3F 0x5E943CB7 0x7E3E229E 0xA705CE1D 0x7DE01AAF 0xEE1B3A15 0xEEF6139F 0x84CFDBAF 0xB6FD75E4 0x374538F7
0xAE33FE7C 0x2C3CAB33 0x76E678FB 0xE5F2099B 0xE2CB7ED7 0x977AA123 0x763BB1A5 0x448E982B 0x584D05B3 0x7E7BCCE3
0xDD1C94C7 0x8A5A9A9F 0xC8D09ABB 0x6D74FD3C 0x00111118 0xEEEEEEEE 0x4D4DAF67 0x97FD6316 0x9D5F776E 0xB150F922
0xD02611B5 0xBF37D82 0xF9EEF977 0x014F9573

[SEROM] Mfg data
0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000 0x00000000

[SEROM] Serial No.
0x00000000 0x00000000

[SEROM] GPIO breadcrumb = 0x00003000
[SEROM] Entering recovery mode...
```

This means you are now back in SEROM. There are two error codes printed error and extended error.

SEROM Error codes are described below.

Extended error codes are for reporting issues (specifically) with the CC312 Crypto Engine.

SEROM Error Codes

// Error codes

```
#define SEROM_STATUS_SUCCESS 0x0
```

// Crypto errors

```
#define SEROM_BSV_INIT_FAIL 0x1
```

```
#define SEROM_BSV_LCS_GET_AND_INIT_FAIL 0x2
```

```
#define SEROM_BSV_LCS_GET_FAIL 0x2
```

```
#define SEROM_BSV_SEC_MODE_SET_FAIL 0x3
```

```
#define SEROM_BSV_PRIV_MODE_SET_FAIL 0x4
```

```
#define SEROM_BSV_CORE_CLK_GATING_ENABLE_FAIL 0x5
```

```
// MRAM errors
#define SEROM_MRAM_INITIALIZATION_FAILURE 0x6
#define SEROM_MRAM_INITIALIZATION_TIMEOUT 0x7
#define SEROM_MRAM_WRITE_FAILURE 0x8

// ATOC errors
#define SEROM_ATOC_EXT_HDR_OFFSET_ZERO 0x9
#define SEROM_ATOC_EXT_HDR_OFFSET_TOO_LARGE 0xA
#define SEROM_ATOC_OBJECT_OFFSET_ZERO 0xB
#define SEROM_ATOC_OBJECT_OFFSET_MISALIGNED 0xC
#define SEROM_ATOC_OBJECT_OFFSET_TOO_LARGE 0xD
#define SEROM_ATOC_OBJECT_OFFSET_TOO_SMALL 0xE
#define SEROM_ATOC_EXT_HDR_OFFSET_MISALIGNED 0xF
#define SEROM_ATOC_HEADER_OFFSET_INVALID 0x10
#define SEROM_ATOC_HEADER_CRC32_ERROR 0x11
#define SEROM_ATOC_HEADER_STRING_INVALID 0x12
#define SEROM_ATOC_NUM_TOC_ENTRIES_INVALID 0x13

// Certificate errors
#define SEROM_CONTENT_CERTIFICATE_NULL 0x14
#define SEROM_CERTIFICATE_NULL 0x15
#define SEROM_CERTIFICATE_CHAIN_INVALID 0x16
#define SEROM_INVALID_OEM_ROT 0x17
#define SEROM_CERTIFICATE_ERROR_BASE 0x18
#define SEROM_CERTIFICATE_1_ERROR 0x19
#define SEROM_CERTIFICATE_2_ERROR 0x1A
#define SEROM_CERTIFICATE_3_ERROR 0x1B

// BOOT errors
#define SEROM_BOOT_CODE_LOAD_ADDR_INVALID 0x1C
#define SEROM_BOOT_VERIFY_IN_MEMORY_CASE_INVALID 0x1D
#define SEROM_BOOT_ZERO_IMAGE_LENGTH_INVALID 0x1E
#define SEROM_BOOT_ENCRYPTED_IMAGE_INVALID 0x1F
#define SEROM_BOOT_VERIFY_IN_FLASH_CASE_INVALID 0x20
#define SEROM_BOOT_IMAGE_LENGTH_TOO_LARGE 0x21
#define SEROM_BOOT_RAW_IMAGE_LOADING_NOT_ALLOWED 0x22
#define SEROM_BOOT_SERAM_JUMP_RETURN_ERROR 0x23
#define SEROM_BOOT_FAILED 0x24
#define SEROM_BOOT_JUMP_ADDRESS_NOT_VALID 0x25

// Bank selection errors
#define SEROM_BOTH_BANKS_INVALID 0x26

#define SEROM_ATOC_EXT_HDR_OFFSET_TOO_SMALL 0x27

#define SEROM_BOOT_END_OF_MAIN_ERROR 0x28
#define SEROM_INVALID_NULL_PTR 0x29
#define SEROM_INVALID_TOC_OFFSET 0x30
```

11. Primary Boot Path Alternative

On ENSEMBLE E4, E6 and E8 devices there is a capability to supplement MRAM as the Primary boot source for the SES and STOC.

The external memory support is Octal SPI. As SEROM is responsible for booting SES (from MRAM) it needs to know the details regarding OSPI settings, these are stored in OTP on the SoC.

If a user wants to enable the use of External memory as a backup to MRAM they will have to program the OTP to enable this mode.

The Maintenance tool is used to create binary objects from a JSON description. These binary files are used via ISP to program the OTP its.

12. CONDUCTOR Tool Flow

The ALIF CONDUCTOR tool is a web-based application that allows a user to configure a Device. Please refer to the CONDUCTOR documentation for more details. This section will focus on the CONDUCTOR material generated for STATIC configuration and how to get this configuration data built using SETOOLS.

CONDUCTOR can be used to generate configurations for

- Static Pin Mux
- Static Clocks
- Static Application space Firewall

The output of the CONDUCTOR tool is a JSON file that can be included into an application as a DEVICE Config object. This object is executed by SES on boot up of the device, so by the time the Application CPUs are booted settings for PinMux, Clock and Firewall are already set.

This method is known as STATIC configuration. As the configuration is stored in Application MRAM space (as part of the ATOC) it will remain there until deleted or overwritten.

Configuration Options

There are two main methods of configuring the device for PinMux, Clock and Firewall:

- STATIC
- DYNAMIC

STATIC configuration

Configuration data is part of the ATOC (DEVICE Configuration object) and will be installed on each boot of the device before the Application cores are booted.

DYNAMIC configuration

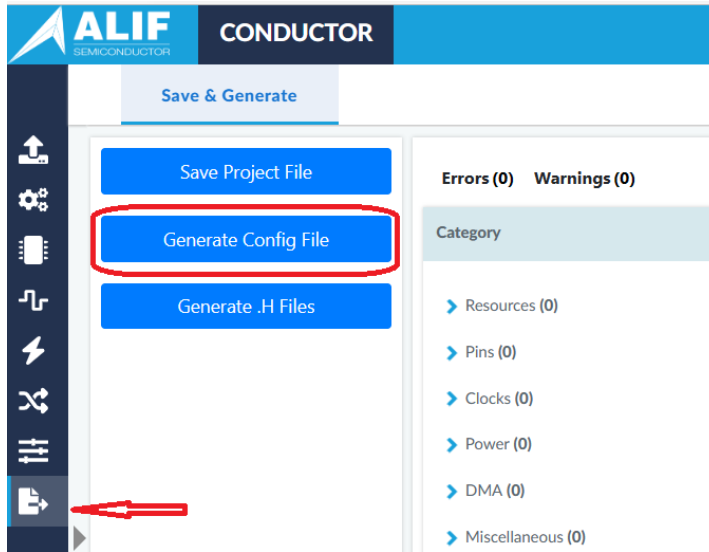
Configuration is done using SERVICE library calls from an application core once booted. This means the initializations are done at run-time or dynamically. These settings can also be changed by subsequent SERVICE library calls.

Using the CONDUCTOR Tool with SETOOLS

The Conductor tool generates two types of output – Device Configuration and C header files. SERAM uses the Device Configuration output. It is a JSON file that needs to be copied to the APP release folder (under build\config) and then referenced by the DEVICE entry in the ATOC config file.

Steps –

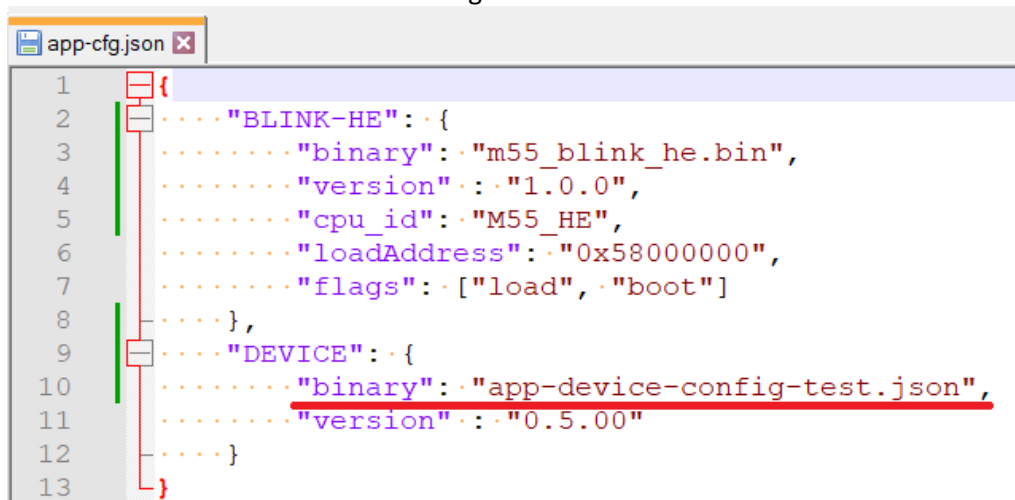
1. Generate a Device Configuration File from the Conductor tool -



The default output file name is something like 'Config__AE722F80F55D5AS_20230705_0910.json'.

You can rename the file at this point or do that later.

2. Copy the file to the <APP release folder>\build\config.
The Conductor tool saves the generated file in the browser's default Downloads folder, so copy it from there to your APP release.
3. Optionally, rename the file, e.g., to 'app-device-config-test.json'.
4. Reference the file from the ATOC config file –



Miscellaneous Feature set supported from CONDUCTOR tool

The MISECELLANEOUS section of the Conductor tool has several configuration options that will be passed into SES for processing at Boot time.

The options that are supported are as follows:

- LFXO_CAP_CTRL
- LFXO_GM_CTRL
- HFXO_CAP_CTRL
- HFXO_PFET_GM_CTRL
- HFXO_NFET_GM_CTRL
- SE_BOOT_INFO

The unsupported options are printed as Warnings when running the app-gen-toc tool

```
[INFO] Create area for: miscellaneous
[INFO] Process Miscellaneous
[INFO] [WARN] SE not supported: ISP_MAINTENANCE_SUPPORT
[INFO] [WARN] SE not supported: FW_RUNTIME_CFG
[INFO] [WARN] SE not supported: PINMUX_RUNTIME_CFG
[INFO] [WARN] SE not supported: CLOCK_RUNTIME_CFG
[INFO] [WARN] SE not supported: BOARD_LED_COUNT
[INFO] [WARN] SE not supported: BOARD_LED_RGB_COUNT
[INFO] [WARN] SE not supported: BOARD_BUTTON_COUNT
[INFO] [WARN] SE not supported: BOARD_CONFIG_JUMPER_COUNT
[INFO] [WARN] SE not supported: BOARD_SWITCH_OUTPUT_COUNT
Calculating APP area...
```

You must use the '-v' (Verbose) option to see these Warnings.

SE_BOOT_INFO

The default for this is '0' which means all SES Prints are enabled on boot up.

Value	Output
0	All prints enabled
1	
2	Suppress all prints

Non-CONDUCTOR Options

There are some features that are not supported directly from the Conductor Tool.

The options that are supported are as follows:

- BOR_THRESHOLD
- BOR_HYSTERESIS
- SHARED_MEM_CTRL
- REG_ALIASING_CTRL
- SE_CLK_SRC_PLL

Brown Out Reset (BOR) hysteresis and threshold configurable

The device boots using safe default boot BOR parameters – minimal non-zero hysteresis (0x1) and zero threshold (0x0).

It is possible to change the BOR parameters during boot via ATOC device configuration. This feature is currently not supported by the Conductor tool, so the BOR settings must be added to the ATOC device configuration manually.

atoc-device-config.json, 'miscellaneous' section –

```

{
  "miscellaneous": [
    {
      "id": "LFXO_CAP_CTRL",
      "value": 8
    },
    {
      "id": "LFXO_CM_CTRL"
    }
  ]
}

```

... BOR parameters added to the section -

```

{
  "id": "BOR_HYSTERESIS",
  "value": 2
},
{
  "id": "BOR_THRESHOLD",
  "value": 1
}

```

SHARED_MEM_CTRL

Share Memory Control register. This setting provides the value to be written to the Shared Memory Control register on boot. This register controls the address at which the SRAM1 memory appears in the global address space – at its original location 0x0800_0000 or right after SRAM0 at 0x02400000, which provides a contiguous memory region that combines SRAM0 and SRAM1. The register also controls the access from certain peripherals (NPU, ISP, JPEG) to certain global memory regions.

```

{
  "id": "SHARED_MEM_CTRL",
  "value": "0x80000000"
}

```

NOTE: The Shared Memory Control register does not exist in FUSION and BALLETO devices, so for them this setting is ignored.

REG_ALIASING_CTRL

Analogue Register Aliasing setting. By default, it is set to enable aliasing.

```

{
  "id": "REG_ALIASING_CTRL",
  "value": "0x00000001"
}

```

NOTE: The Analogue Register Aliasing register does not exist in ENSEMBLE and BALLETO devices, so for them this setting is ignored.

SE_CLK_SRC_PLL

The default for this is '0' which means the SE Clock source is from HFRC not PLL

Value	Clock Source
0	HFRC
1	PLL

Open Firewall configuration

The SW FW releases come with a default device configuration file '**app-device-config.json**' located in '**build\config**' that can be used as an alternative to the Conductor-generated device configuration.

This file configures the firewall in 'open/flat' mode, in which all Masters in the device have full access to all memories and registers located in the EXPMST0 and EXPMST1 address ranges –

- SRAM0, including 'stitched SRAM1' and 'striped view' (0x02000000-0x03ffffff)
- SRAM1 (0x08000000 – 0x0ffffff)
- EXPMST0 (0x40000000-0x5ffffff) – peripherals and M55 TCMs
- EXPMST1 (0x60000000-0x7ffffff) – BOLT devices only, additional memories

Access to the address range 0x80000000-0xffffffff (MRAM and OSPI) is enabled by the SERAM firmware, so it is not included in the device configuration.

The file '**app-device-config.json**' does not include any pin muxing or clocks configuration and has default values in the sections 'metadata' and 'miscellaneous'. If you want to keep these sections as generated by the Conductor, just edit the Conductor file and replace the 'firewall' section in it with the 'firewall' section from the default '**app-device-config.json**'.

13. Device Debug Strategies

The following section describes some debug strategies that can be applied to a device that appears to be not working.

Target Not responding

The most common seen error from the SETOOLS is “Target is not responding”.

This means that the Host machine is unable to communicate with the Target device using the ISP over the SEUART.

Common causes are

- Baud rate
 - The baud rate is not correct for the target device. ENSEMBLE E7,5,3,1 device use 55000 baud rate, ENSEMBLE E8,6,4 use 57600.
 - Check that your SETOOLS tools-config matches your target device. This will set the baud rate appropriately.
 - All ISP SETOOLS allow for overriding the default baud rate using the -b' option.
- Bad ATOC
 - ATOC built with incorrect DEVICE configuration will cause system to crash.
 - This is seen on ENSEMBLE E1 devices where the default DEVICE configuration is set to ENSEMBLE E8.
 - Users must ensure tools-config is set for ENSEMBLE E1 and re-run app-gen-toc to ensure correct DEVICE configuration is enabled.
 - Use HARD maintenance mode to remove the offending ATOC and re-flash corrected version.
- STOP Mode
 - Application has put the device into STOP mode. This means SES is turned off (along with everything else) and so ISP will not respond
 - Use HARD Maintenance mode to recover.

Recovery vs SES Mode

In normal operational mode SES will be running on the device. In rare cases this may not be the case, and your Device will be in Recovery mode. This means that SEROM failed to find a valid SES image to boot.

In recovery mode you will use the Maintenance tool to put back (Recover) an SES image into the device.

In recovery mode you are communicating with SEROM only. The ISP command set for SEROM is limited compared to when talking to SES.

The Maintenance tool upon execution checks what state the Target device i.e. SEROM or SES mode. If it detects SEROM mode, then a limited set of Maintenance commands will be shown.

HARD Maintenance mode

This mode allows the user to override the SE Boot sequence and get access to the Secure Enclave.

Quick Fix guide

Anyone who is troubleshooting a bricked or unresponsive board should do these basic first steps –

- Check the baud rate is correct for the Target device
 - You can always override the default baud rate using the ‘-b’ option
- If your Target application has put the device into a STOP power mode, then the SEUART will not respond (as the device is powered off)
 - Use Hard Maintenance mode to recover. This is because the [RESET] of the device will start the boot process again and you will be able to suppress the Application CPU being launched (This is the essence of Hard Maintenance mode).

- Check if the board is in SEROM or SERAM mode,

If you are not sure what mode you are in, use SETOOLS

```
C:\Windows\System32\cmd.exe - python3 maintenance.py
3 - Get CPU boot info
4 - Device enquiry
5 - Get revision info
6 - Get OTP data
7 - Get MRAM data
8 - Get log data
9 - Get SEROM trace data
10 - Get SERAM trace data
11 - Get power data
12 - Get clock data

Select an option (Enter to return): 4
SEROM Error = 0x24 (SEROM_BOOT_FAILED) Extended Error = 0xf1000015 Maintenance Mode = <None>
```

- SEROM mode – run the SEROM recovery command.

NOTE: Hard Maintenance mode does not make sense in SEROM mode and is not supported.

Recommendation – run the recovery using the option ‘**2. Recovery (No Reset)**’. When the recovery is complete, open a terminal, then reset the board and monitor the output on the SE UART.

If, after Booting, you are still in recovery mode look for extended Error code in Recovery screen and dump SEROM Trace log

- SERAM mode – enter Hard Maintenance mode and then Erase the Application MRAM.

Maintenance -> Option#3 -> Option#3 (or Option#1)

Reporting Issues to ALIF

When reporting issues to ALIF always provide the following information

- Version of SETOOLS used.
- Version of Device used
- Maintenance -> Option#2 Device information -> Option#5 Get Revision Information
- SE-UART output on boot up
 - This is the most useful for triaging issues
- SEROM, SERAM TRACES (Maintenance -> Option#2 Device Information -> Option#9 and Option#10)

14. Life Cycle State SECURE

Some notes on LCS=SE Policy and operation

- In LCS=SE it is not possible to open JTAG debugger access to the Secure Enclave.
- Read access to MRAM and OTP via ISP is disabled, to prevent reading customer code and data via ISP.
- SES always requires that a secure debug request is signed, regardless of whether the request is for secure debug unlocking or for the RMA.
- In LCS=SE the Firewall is ON, so access to most of the device is disabled. To enable access –
 - Use the Conductor tool to assign resources (memories and peripherals) to CPUs
 - Alternatively, use the default device configuration file 'app-device-config.json' that is provided with the SE releases.
 - This file opens access to all device memories and most of the peripherals, for all Masters.

Please see section 'Open Firewall configuration' in chapter 12. Conductor Tool flow for details.

15. Document History

Version	Date	Change Log
0.84.0	October 2023	Web release for Security Toolkit v0.84.0 tools
0.85.0	November 2023	Initial support for MAC OS in SETOOLS
0.87.0	December 2023	Adding assets-gen and app-provision tool details
1.104.0	February 2025	New Maintenance menus when in Recovery mode
1.112.0	June 2026	Additional BL_ERROR codes added